Re-select or Learn? A Supervisory Layer for Rule-Based HVAC Control

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Abstract

Rule-based HVAC controllers remain widely deployed, and a supervisory layer that adapts their exposed parameters without modifying them is a low-cost way to improve them, but it is unclear when such adaptation helps. We study two sources of values for such a layer, each evaluated separately on matched 30-day episodes of the Smart Buildings Challenge simulator with pre-specified tests: re-selection from a library of weather-conditioned profiles, and bounded online estimation and feedback rules. Within the building families the profiles were tuned on, a one-time gate that re-selects daily only after mild starts outperformed persistence-based re-selection by 0.504 percentage points (95% station-cluster interval [0.337, 0.704]; 98 episodes), but a registered replication on 40 new episodes was inconclusive, and the profiles did not transfer to new architectures. On a blind test of unseen architectures, a learned optimal start and a duct static-pressure reset, a few bounded rules with no offline training, reduced the loss of a tuned fixed controller by 2.82% over its 10 composite architectures (95% architecture-cluster interval [1.48, 3.94]), improving 11 of all 12. Unchanged, the layer reduced loss by 5.71% on a later version of the controller and by 3.37% on a trained neural controller, where its pre-specified per-architecture criterion failed; an energy-targeted rule reduced the loss of a model-based controller by 1.88%. On the competition leaderboard the comfort rules gave no gain or a loss. An exact decomposition bounds a rule's benefit by the share of the controller's loss in the term it targets; post hoc, gains followed this share (Spearman $\rho = 0.80$), and five of seven predictions registered from it held. In simulation, these results support targeting supervisory rules to what a controller leaves uncorrected rather than applying them by default.

1 Introduction

Heating, ventilation and air-conditioning (HVAC) systems must balance occupant comfort against energy cost and carbon emissions as conditions change. Model predictive control and reinforcement learning exploit this variability but may need models or long training and can be hard to deploy and transfer across buildings [1, 2]; many installed systems still run rule-based logic.

A middle ground keeps the controller and lets a supervisory layer adapt its parameters during operation, using a small library of weather-conditioned profiles tuned in advance or values estimated online from the building, with feedback, within fixed bounds. With a library, timing matters: re-selecting often tracks the weather but may switch profiles for little benefit, and since profile effects accumulate over days, they must be judged over long horizons.

We ask when profiles should be re-selected, when the layer should learn instead, and when it helps at all; we study the two sources separately and propose no rule that chooses between them. Using a public rule-based controller [3] in the Smart Buildings simulator [4, 5], we compare policies on matched 30-day episodes and test each source on sets frozen in advance: the profiles on their own

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building families, and learning on new architectures and three further controllers. Our contributions are (i) a supervisory layer over an unmodified rule-based controller with two sources of values, profiles, whose *selection* we separate from reassignment *timing*, and online adaptation of building-specific quantities by bounded rules (Section 3); (ii) a matched evaluation of reassignment schedules over three weather years, in which frequent reassignment paid mainly after mild starts, and a one-time gate built on this pattern modestly beat persistence in a pre-specified test (Section 5.1); (iii) out-of-family and blind tests: the frozen profiles did not transfer, while online estimation with feedback did, in pooled gain, on unseen architectures, unchanged on a later version of the controller and on a neural one, and retargeted to energy on a model-based one (Section 5.2); and (iv) an account of when such a layer helps: its gain is the benefit in the term it targets, bounded by that term's share of the controller's loss, minus its cost in the others, which, found post hoc, fits its gains on four controllers; on shared scenes, five of the seven predictions registered for it held (Section 5.3).

2 Related work

Building control. Benchmarks such as BOPTEST [6], Sinergym [7], CityLearn [8] and the Smart Buildings Control Suite [4] make controllers comparable across buildings and weather, the Smart Buildings Challenge [5] scores them on hidden scenarios, and Building2Building [9] targets transfer across buildings. Besides model predictive control (MPC) [1] and reinforcement learning [2], residual policy learning keeps an existing controller and learns a correction to its actions [10, 11]; we keep the controller without learning a policy, adapting its exposed parameters or bounding one command, and ask how much this can add, given what the controller leaves uncorrected (Section 5.3).

Adapting rule-based controllers. Control has long been adapted to conditions, from gain scheduling [12] to supply-temperature reset [13] and optimal start, which learns from past recoveries when to start [14, 15]. Contextual Bayesian optimisation tunes parameters to the context under constraints [16, 17]; our rules are bounded, with no online black-box optimisation. Distribution tuning of rule-based HVAC constants [18] argues for comparing adaptive schemes with tuned static ones. Timing is a design choice too: event-triggered MPC re-optimises only when needed [19], and we compare schedules that re-select profiles from a fixed library on matched episodes. The layer's main rules are a recovery-based optimal start and the duct static-pressure trim-and-respond of Guideline 36 [20]; we contribute evidence on when they help, not new mechanisms. Because choosing configurations or rules after seeing outcomes is optimistic [21], we separate development from frozen evaluation, as in pre-registration [22], and test where a lack of transfer would stay hidden.

3 Method

3.1 Problem formulation

Controller and layer. A stateful rule-based controller maps observations x_t and parameters θ_t to actions $u_t = C(x_t, \theta_t)$; B denotes the fixed baseline configuration. A supervisory layer sets θ_t during operation without modifying C, from one of two sources (Figure 1): a profile library Φ whose entries a schedule π selects at daily decision points (Section 3.2), or a state ξ_t updated from the building's response, with $\theta_t = g(\xi_t) \in \Theta$, where Θ contains only quantities the controller already exposes, within bounds fixed in advance; this source may also floor the static-pressure command of each air handler a, $u_{t,a} \leftarrow \max(u_{t,a}, f_{t,a})$ once requests have raised $f_{t,a}$ above its minimum (Section 3.3). For profiles the weather regime proposes a profile and the schedule decides whether to adopt it; for learning a rule fixed in advance determines whether to act and the building's response how much. At runtime neither source uses future weather or the reward, and the learning state is reset to fixed values every episode.

Objective. The simulator reward is $r_t = -(0.6\,q_t + 0.2\,c_t + 0.2\,e_t)$, with non-negative normalised comfort regret q_t, energy cost c_t and carbon e_t. The loss $\ell(s, \pi)$ of a policy π (a configuration, schedule or layer) in episode s is the mean of $-r_t$ over its $|T(s)| = 8{,}640$ scored steps, and the gain of π over a comparator π' on a matched set S with weights w_s is the reduction in pooled loss

$$G_S(\pi; \pi') = 100 \left(1 - \frac{\sum_{s \in S} w_s\, \ell(s, \pi)}{\sum_{s \in S} w_s\, \ell(s, \pi')} \right). \tag{1}$$

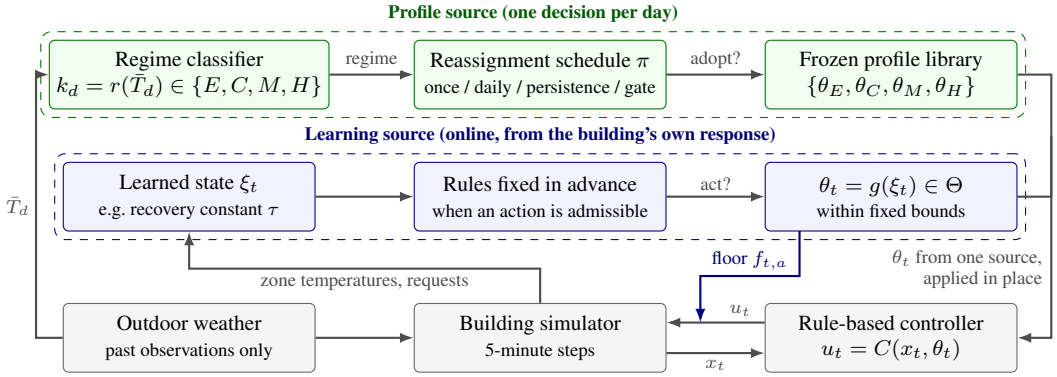


Figure 1: The supervisory layer sets the values θ_t of an unmodified rule-based controller in place, preserving its integrators and the plant state, from a frozen profile that a schedule selects daily from the weather regime (top, Section 3.2) or from a state updated online from the building's response, within fixed bounds, when a rule fixed in advance admits an action (middle, Section 3.3); this source can also floor one command at $f_{t,a}$.

It is not a mean of episode-wise gains. Policies are compared by their difference in gain over B,

$$\Delta_S(\pi_1, \pi_2) = G_S(\pi_1; B) - G_S(\pi_2; B) = 100\,\frac{\sum_{s\in S} w_s\,[\ell(s,\pi_2) - \ell(s,\pi_1)]}{\sum_{s\in S} w_s\,\ell(s,B)}, \tag{2}$$

in percentage points (pp) of baseline-referenced gain. The profile source is evaluated against B, its policies compared by Equation (2); the learning source is evaluated against G30, the tuned configuration it runs on (Section 4), by Equation (1) with equal weights.

3.2 Profile source

Regimes and decisions. The frozen library $\Phi = \{\theta_E, \theta_C, \theta_M, \theta_H\}$ holds one profile per temperature regime. On day d of a 30-day episode a decision point occurs at step $t_d = 228 + 288(d-1)$, and the mean observed outdoor temperature $\bar{T}_d$ over the preceding 228 five-minute steps (19 h) is mapped to a regime $k_d = r(\bar{T}_d)$: extreme-cold (E) if $\bar{T}_d < -15\,^\circ$C, cold (C) if $-15 \le \bar{T}_d < 11\,^\circ$C, mild ($M$) if $11 \le \bar{T}_d < 24.5\,^\circ$C, and hot ($H$) otherwise. At each t_d the schedule π decides whether to replace the active profile by θ_{k_d}. All schedules start under B, and profiles are applied in place, keeping the controller's integrators and the plant state (Algorithm 1 in Appendix F).

Schedules. All adopt the first decision's profile and differ only in later changes: **PS5** makes none (select once), **PD5** may reassign daily, **LIST** and **WEEK** only on predefined days (sparse), **TRIG2** only after a new class appears in two successive complete windows (persistence), and **MG**, the one-time gate, follows PD5 if the first class is mild and PS5 otherwise, a choice it never revisits. Profiles and thresholds were fixed in exploratory development, whose results motivated MG (Section 4).

3.3 Learning source

The learning source runs on G30 (Section 4) and sets the controller's pre-start lead and a floor $f_{t,a}$ under each air handler's static-pressure command. Its rules draw on optimal start [14, 15] and Guideline 36 [20]; rules and bounds were fixed in advance and never tuned on test data (Appendix H).

Learned optimal start. With the occupied comfort band $[T^{\mathrm{lo}}, T^{\mathrm{hi}}]$ and the air temperature $T_{z,t}$ of zone $z \in \mathcal{Z}$, the morning deficit

$$D_t = \frac{1}{|\mathcal{Z}|}\sum_{z\in\mathcal{Z}} \max\big(T^{\mathrm{lo}} - T_{z,t},\ 0,\ T_{z,t} - T^{\mathrm{hi}}\big) \tag{3}$$

covers warm-up and cool-down alike. Under a first-order recovery $D(t+h) = D(t)\,e^{-h/\tau}$ with effective time constant τ, the lead that reaches a residual deficit $D^\star$ at the opening time t^{o} is

$$\lambda_t = \tau \ln \frac{\max(D_t, D^\star)}{D^\star}. \tag{4}$$

Table 1: Evaluation sets: cohorts of the six building families of the profile library and out-of-family sets of new architectures. Counts are not additive within cohorts (arms were added to earlier episodes; stations and variants overlap). Weather: year of the windows, or those of the official training scenarios (development) and one January and one late-June window (held-out). C/M/H: classes by design; observed ones can differ.

Weather	Role	Episodes	Stations	C/M/H	Weights
Profile source: cohorts of the six building families					
2023	exploratory: fixed vs. adaptive policies	70	30	23/24/23	reference masses[a]
2022	frozen schedules, reused stations	58	27	18/23/17	equal
2022	frozen schedules, additional stations	30	13	10/12/8	equal
2021	**pre-specified test (primary)**	98	42	30/42/26	equal
2021	pre-specified optional extension	20	17	6/9/5	equal
2021	registered replication, new episodes	40	28	12/18/10	equal
Learning source: out-of-family sets of new architectures					
training	development (9 architectures)	18	5	4/8/6	equal
Jan., Jun.	held-out (6 architectures)	12	1	6/6/0	equal
2022	**blind test (12 architectures)**	36	28	12/12/12	equal

[a] $(W_C, W_M, W_H) = (574, 814, 467)/1855$: shares of 1,872 starts less 17 extreme-cold (Appendix A).

On weekday mornings, within $t^{\mathrm{o}} - L^{\max} \le t < t^{\mathrm{o}} - L^{\min}$, conditioning starts at the first step with $t^{\mathrm{o}} - t \le \lambda_t$; otherwise the controller keeps its own lead $L^{\min}$. The first hour of each informative recovery updates a smoothed estimate of τ within $[\tau^{\min}, \tau^{\max}]$, and two consecutive mornings with a deficit above its prediction by a margin return the start to the controller for the episode.

Pressure reset and precooling. During conditioning hours a zone warmer than T^{hi} with its cooling loop near saturation requests more air, and each air-handling unit trims its floor $f_{t,a}$ without requests and raises it with them, within fixed bounds (trim-and-respond). For a controller whose loss is mainly energy, a rule of the same form gives learned night precooling: on warm nights with cool outdoor air, the layer lowers the unoccupied cooling limit within bounds if the zone excess before opening, smoothed over earlier mornings without precooling, is positive (Appendix H).

A template for rules. Each mechanism is a bounded rule with five parts: the signal it reads, the condition that triggers it, the command it may change, the bounds of that change, and the condition that returns control to the controller; Table 18 also lists what each needs from the controller. The start and precooling learn; the reset is bounded feedback. A new rule is written by hand for the loss term it targets, and Section 5.3 indicates where it can pay: learned precooling was added this way when the start failed its development gate on Penguin, and was then validated on unseen architectures.

4 Experimental setup

Simulator and episodes. Experiments use the Smart Buildings Challenge simulator [4, 5] at five-minute resolution with historical NOAA weather [23]. An *episode* pairs one generated building with one weather window over a 30-day scoring horizon (8,640 steps, including unoccupied periods), and all policies of a comparison run on the same episodes.

Controllers and comparators. The main controller, K07 [3], is public and rule-based; we use two fixed configurations: the baseline B, tuned during development for a first-workday objective, and G30, a stronger comparator selected from 64 candidates on 12 development episodes for the 30-day horizon. Profile schedules run B for the first 19 h, so their contrasts with fixed policies include a start-up difference; the learning source runs on G30 from step zero and is also tested on other controllers: Penguin [24], which learns a zone model online and plans one step ahead; ZT Best [25], the K07 author's later version (a maintenance test); Residual PPO [26], a rule-based controller with a residual policy trained by reinforcement learning; and a graph network imitating it.

Evaluation sets. Each source was developed on its own sets and tested on sets frozen before they were run (Table 1). The profile source was developed on cohorts of its library's six building families (Appendix A; chronology in Figure 4); the mild-start pattern of the 2022 results motivated the gate,

Table 2: The gate gained most and passed the pre-specified test (98 matched 2021 episodes). Top: gain versus B with 95% station-cluster intervals, not independent across policies (shared episodes); changes: mean post-initial profile changes per episode; *Worse*: share of episodes with over 1% higher loss than B. Bottom: the primary contrast in pp of baseline-referenced gain (Equation (2)); secondary contrasts in Table 14.

Policy	Gain (%)	95% interval	Changes	Worse (%)
B (fixed baseline)	0	—	0	—
G30 (tuned fixed)	+3.24	[+2.46, +4.11]	0	2.0
PS5 (select once)	+3.69	[+2.56, +4.84]	0	10.2
TRIG2 (persistence)	+3.96	[+2.87, +5.20]	2.33	7.1
PD5 (daily)	+4.13	[+3.02, +5.38]	4.72	5.1
MG (one-time gate)	+4.47	[+3.32, +5.71]	2.19	3.1
Contrast	Δ (pp)	95% interval		
MG − TRIG2 (primary)	**+0.504**	[+0.337, +0.704]		

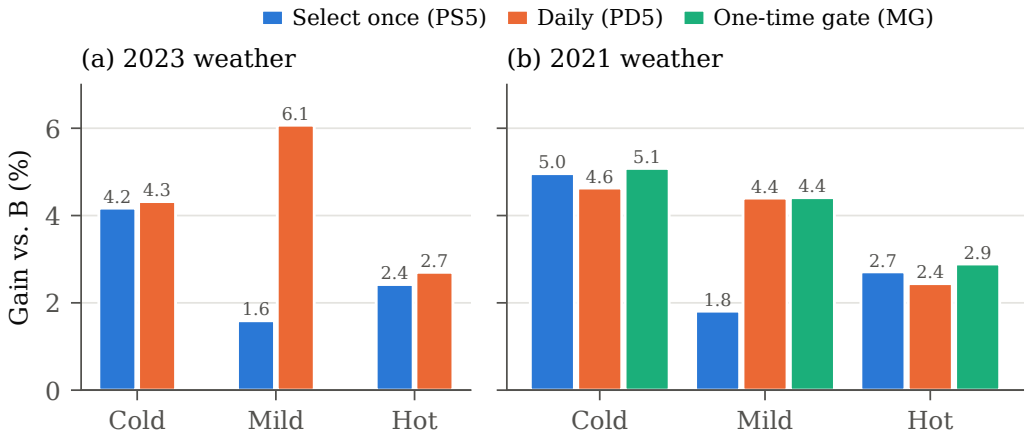


Figure 2: Re-selection pays mainly after mild starts: gain versus B by initial weather class on the library's building families, (a) 2023 weather (reference-weighted within class) and (b) 2021 weather (equal cell weights; classes by design, of which three differ from the observed ones; counts in Table 1).

whose test on new 2021 episodes was fixed in an internal protocol before any 2021 outcome was seen. The learning source was developed on new architectures from the official scenario generator, checked once per mechanism on a held-out set, and tested on a blind set first run after its rules and analyses were frozen; later tests and variants on its scenes are marked as such (Appendices I and J); ten blind architectures have composite floor plans and two out-of-distribution (OOD) shapes. Six official training scenes serve for diagnosis; the Kaggle leaderboard scores hidden scenes.

Statistical analysis. Intervals are 95% percentile intervals from 3,000 cluster-bootstrap resamples of the stations of a cohort or the architectures of an out-of-family set (Appendix E). The single pre-specified primary estimand of the profile source is $\Delta_S(\mathrm{MG}, \mathrm{TRIG2})$ on the 98 main 2021 episodes (Equation (2)), with success defined as a positive estimate whose interval excludes zero; secondary contrasts are Holm-adjusted [27]. Gains of the learning source are pooled over the composite architectures (Equation (1)), with OOD reported separately; exact Wilcoxon tests check the ten blind ones (Appendix J). Each learning mechanism had to pass a pre-specified development gate, and every pre-specified or registered test was frozen in a protocol before its runs (Table 12).

5 Results

5.1 When to re-select profiles?

The gate passed its pre-specified test. On the 98 main 2021 episodes all five alternatives reduced loss relative to B (Table 2), the gate most, by 4.47% against 3.96% for persistence-based reassignment, and the pre-specified primary contrast, $\Delta = +0.504$ pp (95% station-cluster interval $[+0.337, +0.704]$), met its success criterion; the difference is modest, about a ninth of the gate's gain over B. The gate made about as many profile changes as persistence, fewer than half those of daily reassignment, and had the lowest share of episodes more than 1% worse than B among the adaptive schedules. The secondary contrasts agree; those against G30 and between daily and select-once have wide intervals and Holm-adjusted p-values near 0.05 (Table 14).

Re-selection paid mainly after mild starts. On 2021 weather (Figure 2b), daily re-selection beat select-once by 2.59 pp after mild starts (4.40% versus 1.81%) but was slightly worse after cold and hot starts, and the gate, following its fixed mild-start rule, improved on B in every class. This mild-start advantage had appeared in development, on 2023 weather, where daily re-selection was also slightly better after cold and hot starts (Figure 2a), and in both 2022 cohorts (Table 10). By branch, the gate gained by avoiding persistence-triggered switches after cold and hot starts and by reacting faster after mild starts (Appendix E). The explanatory model of Appendix G assumes longer spells after mild starts, which the 2021 weather does not support; post hoc, re-selection gained on cold or hot days that left the mild profile and lost on mild days after cold or hot starts (Table 17).

Sensitivity checks held; replication inconclusive. The primary contrast survived re-weighting, subsets of building variants, another source of class labels and leaving out one station or family (Figure 5). The 20-episode extension was inconclusive (-0.004 pp, post hoc interval $[-0.45, +0.71]$); a registered replication on 40 new episodes gave $+0.255$ pp ($[-0.037, +0.709]$), missing its criterion of a lower bound above zero, so it stays inconclusive even though all 158 episodes, a registered secondary analysis, gave $+0.366$ pp ($[+0.206, +0.535]$; Appendices C and E).

5.2 When to learn instead?

Profiles failed the out-of-family tests. On the out-of-family development architectures the gate lost to G30 by 6.8 pp; on the held-out architectures G30 gained 4.1% over B while the gate lost 0.9%; on the leaderboard it scored worse than B (Table 27). The non-mild profiles change eleven settings of B each, so outside their families they transplant configurations tuned elsewhere.

The learning source gained on unseen architectures. The layer keeps G30 and adapts its start and pressure command online (Table 3). G30 reduced the loss of B by 8.62%, and the layer a further 2.82% (Wilcoxon $p = 0.004$), improving 9 of the 10 composite and both OOD architectures; on the held-out set it gained 3.9%, but only two of four composite architectures improved, so the rule failed (Table 4). Learning the time constant added 0.94% inside the full layer, as registered ($[0.10, 1.82]$, 7 of 10; exact Wilcoxon $p = 0.084$), and 0.95% without the reset (8 of 10; Holm-adjusted $p = 0.023$

Table 3: Pooled gains of the learning source on unseen architectures and other controllers. Blind set (12 unseen architectures × 3 weather windows): gain (Equation (1)) pooled over the 10 composite architectures, with 95% architecture-cluster intervals, bootstrap p-values and the number of composite architectures improved; the two OOD architectures are reported in the text and in Appendix J.

Contrast	Gain (%)	95% interval	p[a]	Better (of 10)
Tuning: G30 vs. B	$+8.62$	$[+6.01, +11.41]$	< 0.001	10
Method: G30 + layer vs. G30 (primary)	$\mathbf{+2.82}$	$[+1.48, +3.94]$	< 0.001	9
Combination: G30 + layer vs. B	$+11.19$	$[+7.78, +14.52]$	< 0.001	10
Learning: learned vs. fixed start, no reset	$+0.95$	$[+0.11, +1.82]$	$0.023^{\dagger}$	8
Pressure reset over learned start	$+1.58$	$[+0.58, +2.48]$	$< 0.001^{\dagger}$	9
Penguin + learned precooling vs. Penguin	$+1.88$	$[+1.12, +3.07]$	$< 0.001^{\dagger}$	10
ZT Best + layer vs. ZT Best	$+5.71$	$[+3.64, +7.55]$	—‡	10
Residual PPO + layer vs. Residual PPO	$+3.37$	$[+1.57, +5.22]$	—‡	8
Graph network + layer vs. graph network	$+3.55$	$[+1.82, +5.40]$	—‡	8

[a] Architecture-cluster bootstrap (approximate with 10 clusters; exact Wilcoxon p in Table 22); †Holm-adjusted over the six secondary contrasts of Appendix J. ‡The layer's pre-specified rule (positive pooled gain, most changed architectures better, none more than 1% worse) held on ZT Best, not on Residual PPO or the graph network.

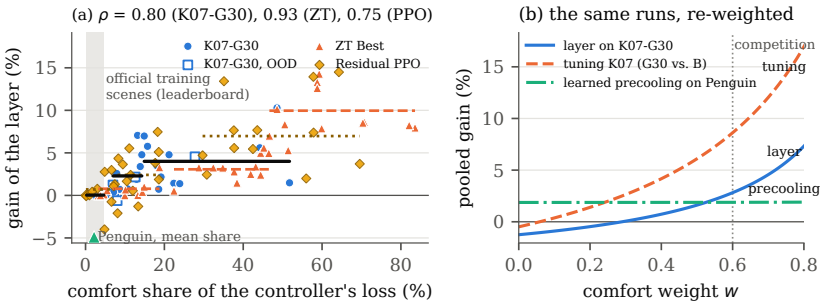


Figure 3: The gain follows the comfort share of the controller's loss (post hoc for K07-G30). (a) Gain of the layer per blind episode against the comfort share of the controller's loss: K07-G30 (circles; squares: OOD) and, in the registered tests, ZT Best (triangles) and Residual PPO (diamonds); segments: pooled gain of each third (K07-G30: $+0.03$, $+2.29$, $+4.00\%$; ZT Best, dashed: $+0.78$, $+3.07$, $+9.96\%$; Residual PPO, dotted: $+0.70$, $+2.41$, $+6.97\%$); grey band: K07-G30 on the official training scenes; green triangle: Penguin's mean share. (b) The same runs re-weighted with comfort weight w; dotted line: competition weights.

by bootstrap, 0.098 by exact Wilcoxon); frozen at 1 h, the start kept only a quarter of its gain. Its advantage built up over two weeks while the pressure reset paid at once (Figure 6).

Other controllers also gained. Penguin's [24] loss is 36% lower than K07-G30's and 97% energy and carbon; the K07 start failed its development gate there, whereas learned night precooling, aimed at energy, reduced its loss by 1.88% and improved all 12 architectures. On ZT Best [25], the K07 author's later version, on the same blind scenes, the unchanged layer gained 5.71% ($[3.64, 7.55]$; all 12 improved), and its pre-specified maintenance rule held. On a trained neural controller [26], itself better than K07-G30, it gained 3.37%, 4.70 points less comfort loss for 1.34 more in energy and carbon, and 3.55% ($[1.82, 5.40]$) on a graph network imitating it, but the rule failed on both, with one or two architectures more than 1% worse. Alone, as registered, the pressure reset gained 2.99% on the neural controller ($[1.60, 4.53]$, all 10 better) and the learned start 0.49%, which caused both losses there (1.35, 1.15%); the reset alone would have passed the rule (Appendix J).

5.3 When does the layer help?

Largest gains on extreme days, losses on holidays. Post hoc, the layer gained 4.1 and 4.2% on the coldest and hottest tenth of days against 2.1% otherwise, and slightly more on Mondays than on other weekdays (3.1 against 2.8%); six of its ten worst days were public holidays, unoccupied in the simulator but unknown to the controller, so its earlier start bought no comfort (Figure 7).

The gain followed the comfort share of the loss (post hoc). Across the 36 blind episodes of G30, Spearman $\rho = 0.80$ ($[0.57, 0.89]$ by architecture; thirds in Figure 3a), and $\rho = 0.78$ across architectures, whereas their shared denominator alone gives a mean ρ of -0.09 when the loss reductions are permuted. By construction, the gain is the share times the part of the targeted loss removed, 15–24% on all three comfort controllers, minus the cost in the other terms (Equation (7), Table 25). The share thus bounds the benefit, and the net gain unless the other terms improve; it measures the headroom rather than predicting the gain, though its first-week value ranked the later gain ($\rho = 0.72$; exploratory, Table 26). On Penguin the mean comfort share is 2.2%, and learned precooling, aimed at energy and carbon, removed 1.82 points without costing comfort. Comfort's 60% weight matters: with equal weights the K07 layer would gain only 0.26% (Figure 3b).

Five of seven registered predictions held. Before the ZT Best runs on the same scenes we registered three predictions: the gain rises with the comfort share (held, $\rho = 0.93$), exceeds G30's gain as ZT Best's mean share does (held: 5.97% against 2.69% on all 36 episodes), and stays within $\pm 0.5\%$ in the lowest-share third (failed: 0.78%). For the neural controller three of four held (the lowest third: 0.70%), and all four for the graph network imitating it, not an independent check (Appendix J).

5.4 The leaderboard and an unoccupied setup

Tuning improved K07's Kaggle score by 2.6%, to 0.01477; the learning source left it unchanged but made the neural controller's 1.0% worse, against a registered prediction of no change (Table 27). A diagnosis of the official training scenes led to a Guideline 36 unoccupied cooling setpoint that passed its development gate (Table 30) and scored 4.3% below G30; on the blind set it lost 0.62%, and with the learned start gained 1.27% over G30 but failed its pre-specified rule. As an untuned third rule of the layer it added 0.38%, with an interval spanning zero ($[-0.45, +1.49]$; Appendix J).

6 Discussion and limitations

Which source, when. Neither source wins by default (Table 28). Profiles paid only on buildings like those they were tuned on; on new buildings a learning rule paid where the controller left loss in the term the rule targets and added little elsewhere. This gives a check that needs no new experiment: before attaching a rule, measure the share of the controller's loss in the term it targets. Unless the other terms improve, the rule cannot gain more than this share, and our comfort rules removed 15–24% of the targeted loss (Section 5.3); on the official training scenes, where K07 and the neural controller leave 0.2–4.6% and 2.5% of their loss in comfort, this leaves about 1% or less, which fits their flat or negative leaderboard results, though the hidden scenes' losses are unknown. The account is post hoc and only partly confirmed by registered predictions, and gains chosen on a leaderboard need not transfer, as the unoccupied setup showed.

Limitations. The two sources were tested on different sets, not head-to-head on matched episodes, and were not combined. The evidence is limited (Table 4): the blind intervals rest on 10 composite architectures from one generator; the layer's pre-specified rule failed on the held-out set, the neural controller and its imitation (Section 5.2); the account of Section 5.3 was tested across controllers on the same scenes rather than on new buildings; the gate was formulated after inspecting earlier results and frozen internally, its extension and its registered replication were inconclusive, its intervals ignore the shared building families, and loss components were not archived for its later cohorts. The layer's value also depends on the objective weights (Section 5.3); it never ran on an untuned controller, and G30 came from a budget-limited search. Longer leads in deep cold triggered the safety freeze in slow buildings; the start rule acts on the mean zone, which can hide the cost in a few zones, and its time constant also absorbs outdoor temperature and season (Appendix J).

Future work. Decomposing the controller's loss by term and hour and attaching the mechanism aimed at the largest term, as we did by hand for Penguin and the unoccupied setup, would systematise transfer; a start rule weighing its night-time cost against the expected comfort deficit would limit the losses above, and stopping it would need a runtime proxy of its gain. Decisions per zone, a holiday calendar and real buildings come next, where bounds and freezes are no safety guarantee.

7 Conclusion

We studied two sources of values for a bounded supervisory layer over an unmodified rule-based HVAC controller: re-selection from a weather-conditioned profile library and online estimation with feedback. Within the library's building families, a one-time gate that re-selects daily only after mild starts modestly beat persistence in a pre-specified test, but its registered replication was inconclusive, and outside those families the frozen profiles did not transfer. On unseen architectures, a learned start with a pressure reset, a few bounded rules with no offline training, reduced the loss of a tuned fixed controller by 2.82%; unchanged, the layer also reduced pooled loss on a later version of the controller and on a trained neural controller, where its per-architecture criterion failed, and an energy-targeted rule helped a model-based controller. The gains are modest and not universal: on the competition leaderboard the comfort rules gave no gain or a loss. An exact decomposition bounds a rule's benefit by the share of the controller's loss in the term it targets; post hoc, gains followed this share, and registered predictions on shared scenes partly confirmed it. This points to selective, controller-aware supervision: measuring where a controller loses before attaching a rule. The study gives no rule for choosing between the two sources and no safety guarantee, and all results come from simulation. Our code, episode definitions and per-episode results are available at `https://github.com/AsteriosPanoras/supervisory-layer-hvac`.

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Use of large language models

Large language models assisted with code development, with running and monitoring experiments, with checking numerical results against their sources, and with drafting, editing and reviewing the manuscript. No LLM was part of any evaluated controller or generated its control actions. Reported simulation metrics and uncertainty estimates were computed by analysis scripts from recorded per-episode results; leaderboard scores came from the competition platform. The authors reviewed the AI-assisted outputs and take full responsibility for the accuracy, originality and integrity of the work.

Overview of the appendices

Appendices A–G document the profile source, H–J the learning source and K compute, software and licences. Table 4 lists what each test showed and its main limit.

Table 4: What each test showed and its main limit. Status: P pre-specified test; R outcome registered before it was known; H post hoc; E exploratory; D descriptive. Gains are pooled over the episodes of the set; pp: percentage points of baseline-referenced gain.

Test	Status	Set	Result	Main limit
When to re-select profiles? (Section 5.1)				
One-time gate against persistence (primary)	P	2021, 98 episodes	$+0.504$ pp $[+0.337, +0.704]$, pre-specified	the 20-episode extension gave -0.004 pp, one of its episodes holding 22.5% of its loss; families not resampled; a registered replication on 40 new episodes gave $+0.255$ pp $[-0.037, +0.709]$, just short of its criterion; all 158 episodes $+0.366$ pp $[+0.206, +0.535]$
Direction of the profile mismatch	H	2021, 98 + 20 episodes	re-selection gained leaving the mild profile ($+11.1\%$ on cold, $+3.5\%$ on hot days) and lost entering it (-6.5%, -2.0%)	post hoc, on the episodes that tested the gate
When to learn instead? (Section 5.2)				
Frozen profiles outside their families	P, D	development, held-out, leaderboard	gate 6.8 pp below G30; -0.9% where G30 gained 4.1%; worse than B	not run on the blind set
Learning on unseen architectures	P	blind, 12 architectures	$+2.82\%$ over G30; 9 of 10 composite and both OOD improved	held-out: $+3.9\%$ pooled but two of four composite improved (rule failed); one generator
The same layer on a later controller	P, R	ZT Best, blind	$+5.71\%$ $[3.64, 7.55]$; all 12 improved	one further controller, with 41.6% higher loss than K07-G30
Learning retargeted to energy	P	Penguin, blind	$+1.88\%$ $[+1.12, +3.07]$; all 12 improved	a different mechanism (the start failed its gate)
The same layer on a trained neural controller	R	Residual PPO, blind	$+3.37\%$ $[+1.57, +5.22]$; 8 of 10 composite improved; three of four registered predictions held	the pre-specified rule failed (two architectures lost 1.15 and 1.35%)

Table 4, continued

Test	Status	Set	Result	Main limit
The same layer on a graph network imitating the neural controller	R	blind	$+3.55\%$ $[+1.82, +5.40]$; 8 of 10 composite improved; all four registered predictions held	imitates the neural controller (validation $R^2 \geq 0.97$); the pre-specified rule failed (52209 lost 1.17%)
Each rule alone on the neural controller	R	Residual PPO, blind	pressure reset $+2.99\%$ $[+1.60, +4.53]$, all 10 better; learned start $+0.49\%$ $[-0.10, +1.03]$; all four registered predictions held	both losses of the layer came with the start; the new arms ran on another machine
Learning inside the full layer: learned against fixed time constant, both with the pressure reset	R	blind, K07-G30	$+0.94\%$ $[+0.10, +1.82]$; 7 of 10 composite better	exact Wilcoxon $p = 0.084$: stronger for the pooled gain than per architecture
Knowing the time constant from day 1 (oracle)	R	blind, K07-G30	$+0.02\%$ $[-0.003, +0.057]$ over the layer, mostly in days 1–14	the value of knowing τ early, not a general upper bound; not deployable
When does the layer help? (Sections 5.3 and 5.4)				
Account of the gain (headroom)	H, R	blind; ZT registered	$\rho = 0.80$, not an artefact of the shared denominator; five of seven predictions registered for two controllers held	post hoc; weak for Penguin across architectures
Time to value; early indication	H, E	blind, 30 composite episodes	learned start ahead of the fixed one from day 6; first-week comfort share ranked the later gain ($\rho = 0.72$)	post hoc and exploratory; not a selector
A third rule: the unoccupied setup added to the layer	R	blind, K07-G30	$+0.38\%$ $[-0.45, +1.49]$; energy and carbon down, comfort up	not tuned for the combination
Unoccupied setup (from a diagnosis of the official scenes)	P, R	development, official scenes, blind	passed its gate; leaderboard 4.3% below G30; on the blind set -0.62% alone, $+1.27\%$ with the learned start	the pre-specified rule failed (one architecture lost 3.3%); the official scenes informed the design
Leaderboard	D, R	hidden scenes	learning unchanged on K07 and 1.0% worse on the neural controller (registered: no change); Penguin with precooling nearly unchanged; the unoccupied setup 4.3% better; the gate worse than B	official scenes' comfort share 0.2–4.6%; one set of scenes

A Cohorts, weather references and study lineage

Common design. An episode pairs one generated building variant with one historical weather window and a declared scoring horizon; matched policies share these inputs. Repeated windows, stations and parent-building families create dependence, so cohorts are analysed separately and their sample sizes are not additive; the 2021 main, extension and replication episodes are distinct but share stations and families, and they are pooled only in a registered secondary analysis (Appendix E). Initial weather strata use the file-derived mean temperature over the first 19 hours and the thresholds of Section 3.2; the class observed by the running policy, from its first 228 five-minute temperatures, can differ from this design stratum. An early three-class reference (1,836 eligible Sunday starts at 36 stations) grouped all temperatures below 11 °C as cold; the later four-class reference (1,872 starts) separates extreme cold (Table 5). Eligibility required at least 72 hourly observations and did not guarantee that a window could support a full 30-day rollout. The 2023 analysis weights each stratum by its reference mass, distributed equally among the sampled episodes of that stratum; the 2022 and 2021 analyses use equal cell weights in the pooled losses.

What differs between studies. Table 6 lists each study with what sets it apart, and Figure 4 gives their order. The first-workday and five-day studies established the baseline and the profiles and

Table 5: Initial-weather reference counts and the 2023 evaluation sample. Reference counts are eligible source windows, not evaluated episodes.

Initial class	Three-class reference	Four-class reference	2023 episodes
Extreme cold (E)	included in C	17	0
Cold (C)	569	574	23
Mild (M)	801	814	24
Hot (H)	466	467	23
Total	1,836	1,872	70

exposed weather-dependent failures; they are not independent 30-day validations. The 2021 test used hourly NOAA ISD weather for 2021 [23], a year not used before. Its frozen window screen required a 720-hour window, starts at least 35 days apart at a station and at most three windows per station; its 42 stations were 30 from the earlier pool and 12 from the additional-station 2022 cohort. The gate, six arms, episode order, primary contrast and analysis were frozen in an internal, time-stamped protocol before any 2021 rewards were inspected; a later amendment changed only retry handling after initialisation failures.

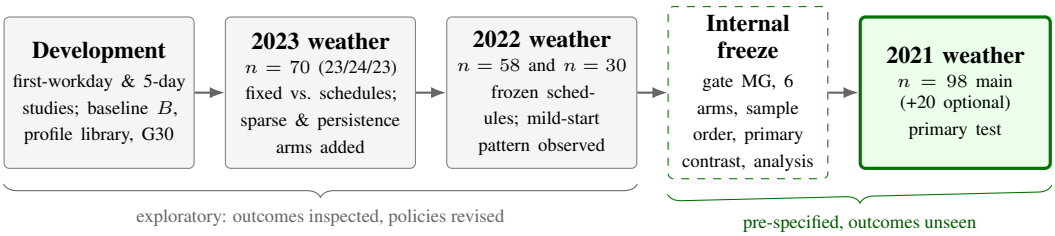


Figure 4: Study chronology. Exploratory stages (grey) informed the profile library and, after the 2022 results, the one-time gate. The gate, six policy arms, episode order, primary contrast and analysis were then frozen before the 2021 evaluation (green). A replication registered before its runs then added 40 new 2021 episodes (Appendix E).

B Reward components and stratum contributions in earlier studies

The later 30-day cohorts archived total loss but not complete component series. Component evidence is available only for two earlier first-workday studies, which used a shorter horizon and selected offline, by the initial weather label, among separately recorded fixed-policy trajectories. Because alternative profiles can change settings that act before the first decision, this is not equivalent to a causal controller that starts under B and then switches; the 30-day cohorts were executed in closed loop. For a cohort with strata k, reference masses F_k and within-stratum mean components $\bar{q}_{a,k}, \bar{c}_{a,k}, \bar{e}_{a,k}$ for policy a,

$$L_a = \sum_k F_k \,(0.6\,\bar{q}_{a,k} + 0.2\,\bar{c}_{a,k} + 0.2\,\bar{e}_{a,k}), \qquad G_{a/B} = 100\,(1 - L_a/L_B). \tag{5}$$

A common normalisation of the reference masses cancels in this ratio.

Early two-action offline selection. On 240 first-workday episodes, a rule selecting B below 11 °C and an alternative configuration otherwise reduced aggregate loss by 1.21%, with energy and carbon terms down 1.69% and 1.94%, but increased reference-weighted comfort regret by 15.99%: unchanged after cold starts, up by 116% after mild and 479% after hot ones, from small bases. Regret increased in 72 episodes, decreased in 16 and was unchanged in 152; the 16 episodes with more than 1% higher loss (3 mild, 13 hot) all had higher regret.

Four-regime offline selection. On 202 first-workday episodes (E/C/M/H = 67/47/48/40), selecting the profile by four-class weather reduced loss by 5.46%, energy cost by 4.77%, carbon by 5.86% and comfort regret by 8.14%. Mean regret fell by 10, 15 and 25% in the extreme-cold, cold and hot strata, but in the mild stratum it increased by 121% from a small base despite lower total loss.

Table 6: Study lineage: what sets each study apart. Added arms, nested horizons and reused windows do not create new independent samples. C/M/H: cold/mild/hot by design.

Study	Episodes	What sets it apart
Early first-workday screen	240	120 windows at 36 stations, each paired with two of 56 building variants; non-random selection: spring under-represented, temperatures near 11 °C excluded
Later first-workday selection	202	204 planned, two all-arm failures; weather-based selection reconstructed offline from recorded static-policy outcomes, not executed as a multi-day controller
Direct five-day / nested 30-day	80 / 36	the 36 with enough weather for a nested 30-day evaluation belong to the 80
2023 30-day comparison	70	72 planned in two tranches of 36; one cold and one hot case of the second failed to initialise in all arms; 70 distinct windows at 30 stations; five arms at first (B, an alternative fixed profile, select-once, daily, predefined-day)
Additional fixed / sparse arms	70 (reused)	four fixed profiles and two sparse schedules on the same 70 episodes, eleven arms in all; in 11 hot-start episodes the original and added arms ran on different machines without demonstrated numerical parity, the other 59 on one machine each
Extreme-cold diagnostics	34 / 13	five-day cases; the 13 longer cases reuse earlier windows
2022 cohorts	58 / 30	previously unevaluated 2022 weather; new variants of the same six families; six arms (B, two fixed alternatives including G30, select-once, daily, persistence); 27 previously used / 13 additional stations; the first ran 60 episodes, and two pilots whose outcomes had been examined were excluded
2022 post-outcome cohort	30	specified afterwards; exploratory; 20 stations
2021 main / optional	98 / 20	pre-specified test at 42 / 17 stations; 100 main planned, half on variants seen under 2022 weather and half on new ones; two new variants failed in all six arms and were excluded without replacement (50 reused, 48 new); C/M/H 30/42/26 and 6/9/5
2021 registered replication	40	registered before its runs (Table 12); arms B, TRIG2 and MG; 28 stations; C/M/H 12/18/10; no episode overlaps an earlier one (Appendix E)

Nine episodes exceeded the 1% loss-deterioration threshold (seven extreme-cold, two mild), all with increased regret.

Contribution by stratum (2023 cohort). Without component series, the 2023 gain can still be decomposed by initial stratum. With $S_{B,k} = 100F_kL_{B,k}/L_B$ the baseline-loss share and $D_k = 100F_k(L_{B,k} - L_{\mathrm{daily},k})/L_B$ the contribution to daily re-selection's gain, cold episodes contribute most (2.05 of the 4.36 pp; mild 1.60, hot 0.71) because they carry the largest share of baseline loss (47%, against 26% each for mild and hot, at reference masses of 31, 44 and 25%). This is a contribution to the gain over B; the advantage of daily re-selection over select-once, a different comparison, is concentrated in mild episodes (Appendix C).

C Extended comparisons and development evidence

2023 cohort: fixed and sparse policies. Table 7 reports all policies on the 70 episodes and Table 8 the gains by initial class. Persistence retained most of daily re-selection's gain (4.24% against 4.36%) with about half the changes (1.99 against 3.93 per episode); predefined-day and weekly schedules also improved on B with fewer changes. GL, the strongest of the originally tested fixed profiles, and G30 both came close to the adaptive schedules. In mild-start episodes, daily re-selection exceeded select-once by 4.48 pp, compared with 0.15 pp in cold and 0.28 pp in hot starts. Two development profiles derived for cold weather, INC and WS1, improved loss in cold-start episodes but incurred about 52% higher loss than B in hot-start episodes (Table 9; HOTINC was derived for hot weather). These selected failures of a mismatched regime do not estimate an unconditional failure rate or test transfer to new buildings, but they show why profile choice must be conditioned on the regime. In 11 hot-start episodes the original and added arms ran on different machines without demonstrated numerical parity (Table 6).

Table 7: 2023 cohort ($n = 70$, reference-weighted). Changes: post-initial profile changes per episode, from the logs of 2,100 daily windows per policy (in total PD5 275, LIST 117, WEEK 71, TRIG2 139).

Policy	Gain (%)	Changes
B	0.00	0
A1 (alternative fixed)	-3.48	0
GL (earlier tuned fixed)	$+3.53$	0
G30 (30-day tuned fixed)	$+3.60$	0
PS5 (select once)	$+3.03$	0
PD5 (daily)	$+4.36$	3.93
LIST (predefined days)	$+4.01$	1.67
WEEK (weekly)	$+3.91$	1.01
TRIG2 (persistence)	$+4.24$	1.99

Table 8: 2023 cohort: gain (%) by initial class. The overall gains of Table 7 weight the classes by their shares of the baseline loss (Appendix B), not equally.

Class (n)	PS5	PD5	GL	WEEK	TRIG2
Cold (23)	4.17	4.32	4.08	3.97	4.26
Mild (24)	1.59	6.07	2.79	5.06	5.76
Hot (23)	2.42	2.70	3.28	2.63	2.66

Table 9: Fixed profiles across initial weather classes: gain (%) versus B (selected cross-regime failures).

Profile	Cold	Mild	Hot	All
INC (cold)	4.15	2.53	-51.84	-10.93
WS1 (cold)	4.61	2.05	-51.79	-10.82
HOTINC (hot)	0.40	-0.97	2.42	0.56

2022 cohorts. Table 10 gives the two 2022 cohorts of frozen schedules. In mild starts, daily reselection gained 4.92% against 1.54% for select-once at previously used stations and 4.57% against -0.92% at additional stations; this mild-start pattern was identified after these outcomes were examined.

Table 10: 2022 cohorts of frozen schedules (equal cell weights; 95% station-cluster intervals).

Cohort	Daily vs. B (%)	Daily $-$ once (pp)
Previously used stations ($n = 58$)	$+4.29$ $[+3.32, +5.42]$	$+0.56$ $[+0.06, +1.18]$
Additional stations ($n = 30$)	$+3.25$ $[+1.88, +4.75]$	$+1.49$ $[+0.17, +3.02]$

Context–parameter associations. During development, high and low values of nine controller parameters around an early configuration were compared on 24 first-workday episodes. Six directional hypotheses linking parameter preferences to weather or building properties were then specified and tested on 24 new episodes with family-centred Spearman correlations, 20,000 within-family permutations and Benjamini–Hochberg control [28] at $q = 0.10$. Three of the four hypotheses on the 19-hour temperature met this exploratory criterion; the fourth and both wall-density hypotheses did not (Table 11). These associations informed development; they do not validate the profiles.

Table 11: Development-stage association tests (ρ: initial and replication samples; p and q_{BH} from the replication; pass: $q_{\mathrm{BH}} \leq 0.10$). T_{19}: mean outdoor temperature over the first 19 h; wall density: density of the interior-wall material in the building's configuration.

Context × parameter	ρ_1	ρ_2	p	q_{BH}	Pass
T_{19} × PRESTART_H	$-.40$	$-.42$	.0357	.0714	yes
T_{19} × EVENING_STOP_H	$-.32$	$-.46$	.0249	.0714	yes
T_{19} × MILD_SHIFT_K	$-.30$	$-.74$	.0001	.0006	yes
T_{19} × HEAT_U_SCALE	$-.28$	$-.15$	.2620	.3930	no
Wall density × KI	$+.40$	$-.16$	.7404	.7404	no
Wall density × EVENING_STOP_H	$-.05$	$-.06$	.5939	.7127	no

Exploratory threshold search. An offline analysis reusing the 2023 fixed-arm outcomes cross-validated one to four temperature partitions and chose a fixed profile per partition. A three-segment fit (cuts near 1.5 and 21.6 °C) gave a cross-validated gain of 5.17% versus B, compared with 2.28% for the hand-set partition under the same procedure. Because the number of segments was chosen after inspecting the scores and the episodes were reused, the 5.17% is exploratory and potentially selection-biased; it does not establish better thresholds for a prospectively evaluated, closed-loop selector.

D Execution and quality assurance

Decision rule. The selector uses $\bar{T}_d$ at step t_d as defined in Section 3.2; the first decision falls on Monday 01:00. If the window lacks any of its 228 observations, the current profile is retained; at the first decision this leaves B active. The controller also retains its own extreme-cold logic based on an exponential moving average of outdoor air temperature, which can respond at a different time from the external selector.

Profile application. At an adoption, all parameters that any profile may override are set to the target profile's values or to the baseline/constructor defaults. A persistent internal attribute storing the pre-start time is synchronised with the pre-start parameter; an early pilot showed that without this step the controller could revert to an obsolete setting (25 of 126 checks failed before the fix; 189 of 189 passed after it, with bitwise parity over 480 steps on nine pilot cases). This restoration is deterministic but is not a proof of bumpless transfer.

2021 gate execution and QA. In the 98 main episodes, 43 entered the mild (daily) branch and 55 the select-once branch; three observed classes differed from the design labels. All 98 gate losses equal exactly the loss of the corresponding PD5 or PS5 arm (maximum relative difference zero). Before inspecting 2021 rewards, six previously evaluated 2022 episodes (two per regime) were re-run: the gate reproduced its branch in rewards, actions, decisions and logs on two local hosts. The hosted runtime was not bitwise identical; a pre-declared tolerance screen admitted it (on the six re-run episodes the gate had to match the corresponding PD5 or PS5 result within 0.5%, with a spread across machines of at most 2%; mean hosted-minus-local difference in the primary contrast -0.024 pp, maximum absolute 0.270 pp). By environment, the primary contrast was $+0.482$ pp on local host A (54 episodes), $+0.633$ pp on the hosted runtime (24) and $+0.379$ pp on local host B (20); because different episodes ran on each, these are not paired hardware comparisons.

Registration record. Every registered test was frozen in a protocol before its runs (Table 12). The repository's integrity folder holds the append-only ledger in which each protocol's SHA-256 was recorded before the outcome it concerns, a map from each published copy (working notes and local paths removed) to the hash of its original, and OpenTimestamps proofs for the registrations from 11 November evening onwards (OTS). The ledger records internal freezes; only the registrations marked OTS also carry independent timestamp evidence.

Table 12: Registration record: each registered test, its protocol in the repository, when it was frozen (EET, as written in the protocol; day only where the protocol gives none) and the first 12 hexadecimal digits of the SHA-256 of the original protocol. OTS: also stamped with OpenTimestamps.

Test	Protocol (`experiments/`)	Frozen	SHA-256	OTS
One-time gate, 2021	`01_profile_study_2021/PROTOCOL.md`	8 Nov 02:59	`9a566eb3859f`	
Learned start, development gate	`02_development_gates/k07_learned_start/PROTOCOL.md`	10 Nov 12:49	`f43b77bfa249`	
Pressure reset, development gate	`02_development_gates/k07_pressure_reset/PROTOCOL.md`	10 Nov	`5bfc700c7ba0`	
Layer on unseen architectures	`04_blind_set_k07/PROTOCOL.md`	10 Nov 15:46	`51e00be89a35`	
B and fixed start on the same scenes	`03_out_of_family_and_held_out/layer_held_out/PROTOCOL.md`	11 Nov 01:30	`fb1d5273fc5e`	
Penguin with learned precooling	`06_penguin_precooling/PROTOCOL.md`	11 Nov 00:47	`65184e4aa16f`	
Unoccupied setup	`08_unoccupied_setup/PROTOCOL.md`	11 Nov 10:26	`9e20135bf6a3`	
ZT Best and its headroom predictions	`05_maintenance_zt_best/PROTOCOL.md`	11 Nov 11:05	`c96cb145ddac`	
Residual PPO and its predictions	`07_neural_controller/PROTOCOL.md`	11 Nov 21:02	`2c6a414e36a6`	OTS
Graph network	`09_graph_network_controller/PREREG_GNNSUP_v1.md`	12 Nov	`317beada4d12`	OTS
Oracle start, third rule, rule ablation	`10_extension_ablation_oracle_rules/PROTOCOL.md`	12 Nov	`af3a09532096`	OTS
Replication of the 2021 gate test	`12_replication_2021/PROTOCOL.md`	12 Nov 12:57	`250d354a3f67`	OTS
Fixed time constant inside the layer	`13_learning_inside_the_layer/PROTOCOL.md`	12 Nov 13:45	`d27dabbad43b`	OTS

E Inference details and sensitivity

2021 primary analysis. The estimand is the pooled-loss gain of Equation (2) with S the 98 main episodes and equal cell weights. The 42 station identities are resampled with replacement 3,000 times (fixed seed), retaining all episodes and arms of each sampled station; the 2.5th and 97.5th percentiles form the interval. The analysis uses the class observed at the first decision, which differs from the weather-file label in three episodes. Table 13 collects the sensitivity analyses.

Primary contrast and regime classes: further detail. Expressed as a direct loss reduction between the two schedules, the primary contrast corresponds to 0.525%; the gate had lower loss than persistence in 58 episodes, higher loss in 15 and identical loss in 25. Decomposed by the branch actually taken, the 55 non-mild episodes (following select-once) contributed $+0.339$ pp and the 43 mild episodes (following daily) $+0.165$ pp; within each branch the gate exceeded persistence by $+0.465$ and $+0.609$ pp respectively, and the non-mild branch contributes more in aggregate because it carries a larger share of baseline loss. On 2021 weather, with the weather-file label as the class, the gains of select-once, daily re-selection and the gate versus B were 4.96, 4.63 and 5.08% after cold starts, 1.81, 4.40 and 4.41% after mild starts, and 2.71, 2.44 and 2.89% after hot starts.

Robustness. The primary contrast survived re-weighting by historical regime frequencies, restriction to subsets of building variants, another source of class labels and leaving out one station or family (Table 13). These checks show how much the estimate depends on single stations or families; they do not model the dependence between variants of a family.

The 20-episode extension (post hoc reading). The pre-specified 20-episode extension gave -0.004 pp; Table 13 also gives it pooled with the main sample, as the protocol reports it. Read after its outcome, it had the main sample's mix of starts (9 of its 20 episodes started mild, as did 43 of the 98) but one dominant episode: a cold January start in Kansas City held 22.5% of its baseline loss, and there the gate trailed persistence by 1.26 pp. Per episode the gate gained in 10 of the 20, lost in 7 and tied in 3; its mean episode-wise gain, a different estimand that averages each episode's own percentage gain and is not a re-weighting of the primary one, was $+0.314$ pp $[-0.098, +0.887]$, against $+0.627$ pp in the main sample. The primary estimand is a pooled-loss gain, a ratio of losses summed over episodes: every episode counts once, but an episode with a larger baseline loss weighs more in the percentage. Its loss-weighted effective number of episodes, $n_{\text{eff}} = (\sum_e L_e^B)^2 / \sum_e (L_e^B)^2$, is 54.5 of the 98 main episodes but only 9.8 of the 20 extension episodes; it measures how concentrated the baseline loss is, not a sample size adjusted for shared stations or families. The extension's interval, computed post hoc in the same way as the primary one, contains the main estimate, so the extension neither confirms nor contradicts it: twenty episodes with one dominant episode are compatible with a modest effect and with none. The effect is modest and not uniform.

Registered replication. A replication registered before its runs (Table 12), on 40 new 2021 episodes at 28 stations, none overlapping an earlier episode, gave $+0.255$ pp (the gate 4.06% and persistence 3.81% over B; interval in Table 13). Its estimate is positive and 95% of its bootstrap resamples lie above zero (post hoc), but its interval just includes zero, so it falls just short of the registered criterion for replication, a lower bound above zero, and far from the criterion for failure, an upper bound below $+0.25$ pp; by its rule it is inconclusive. Its mean episode-wise gain was $+0.517$ pp $[+0.076, +1.125]$ (the gate better in 16 episodes, worse in 6, tied in 18), and no episode held more than 12.2% of its baseline loss (without that one, $+0.291$ pp). Pooled over all 158 episodes of the main sample, the extension and the replication (42 stations), a secondary analysis registered with it, the gate gained $+0.366$ pp (Table 13).

Earlier cohorts. The 2023 cohort resamples complete weather windows within initial class (3,000 replicates), preserving all paired arms of a window; it does not resample stations or families or adjust for multiplicity. Table 15 gives selected contrasts. The daily and select-once gains of 4.36% and 3.03% differ by about 1.33 pp; no paired interval for this contrast was pre-specified for the 2023 cohort.

First-workday and five-day studies. The first contextual study used a cluster-robust variance accounting for crossed building–window and building–station dependence [29], inflated by a calibration multiplier and enveloped by a pigeonhole bootstrap [30]. On the first workday the rule

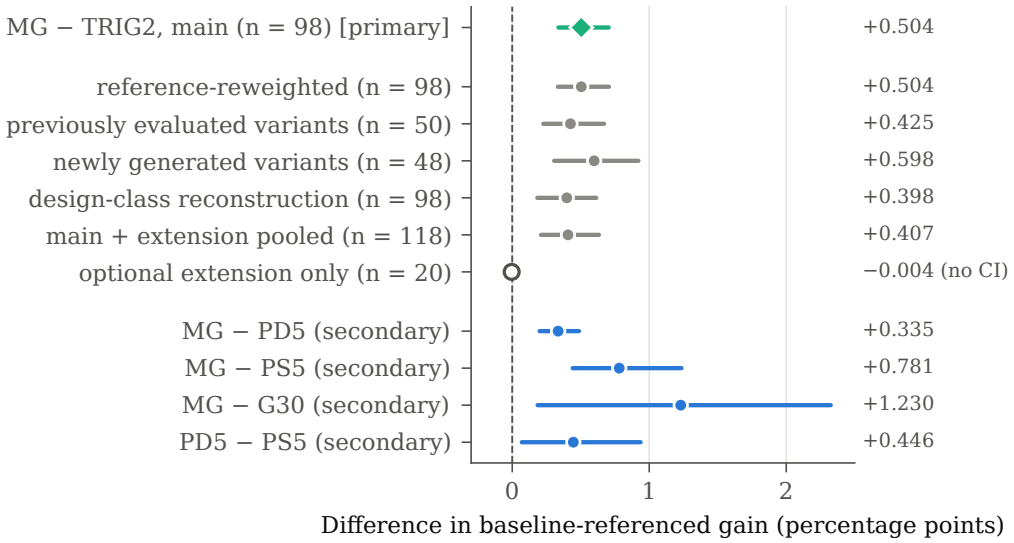


Figure 5: Paired differences in baseline-referenced gain on 2021 weather with 95% station-cluster intervals. Top: pre-specified primary contrast. Middle: sensitivity analyses of the same contrast (the 20-episode extension is shown without its post hoc interval, given in Table 13). Bottom: secondary contrasts (Holm-adjusted p-values in Table 14).

Table 13: 2021 primary contrast (MG − TRIG2, pp): main estimate and sensitivity analyses.

Analysis	n	Δ (pp)	95% interval
Primary: pooled-loss gain	98	$+0.504$	$[+0.337, +0.704]$
Mean episode-wise gain (post hoc)	98	$+0.627$	$[+0.414, +0.841]$
Reference-mass re-weighting	98	$+0.504$	$[+0.333, +0.706]$
Previously evaluated building variants	50	$+0.425$	$[+0.227, +0.671]$
Newly generated building variants	48	$+0.598$	$[+0.306, +0.921]$
Gate reconstructed from design labels	98	$+0.398$	$[+0.183, +0.613]$
Optional extension only (interval post hoc)	20	-0.004	$[-0.451, +0.705]$
Extension without its largest episode (post hoc)	19	$+0.359$	—
Main and extension pooled	118	$+0.407$	$[+0.210, +0.633]$
Main and extension, mean episode-wise gain (post hoc)	118	$+0.574$	$[+0.379, +0.779]$
Registered replication, new episodes	40	$+0.255$	$[-0.037, +0.709]$
All 158 episodes pooled (registered secondary)	158	$+0.366$	$[+0.206, +0.535]$
Leave-one-station-out (range)	—	$+0.462$ to $+0.544$	
Leave-one-family-out (range)	—	$+0.413$ to $+0.591$	

Table 14: Secondary paired contrasts on the 98 main 2021 episodes (pp of baseline-referenced gain).

Contrast	Δ (pp)	95% interval	p_{Holm} [a]
MG − PD5	$+0.335$	$[+0.200, +0.487]$	< 0.001
MG − PS5	$+0.781$	$[+0.441, +1.236]$	< 0.001
MG − G30	$+1.230$	$[+0.185, +2.325]$	0.044
PD5 − PS5	$+0.446$	$[+0.070, +0.937]$	0.044

[a] Holm adjustment over the four secondary contrasts; two-sided p-values computed directly from the 3,000 station-cluster bootstrap replicates. The intervals are not adjusted for multiplicity.

Table 15: 2023 cohort: selected contrasts (pp of baseline-referenced gain).

Contrast	Δ (pp)	95% interval
Daily − GL	$+0.83$	$[-0.40, +2.18]$
Daily − class-wise fixed oracle	$+0.57$	$[-0.19, +1.36]$
Daily − G30	$+0.755$	$[-0.494, +2.132]$
Weekly − daily	-0.45	$[-0.77, -0.21]$
Persistence − daily	-0.12	$[-0.25, -0.003]$
Predefined-day − daily	-0.34	$[-0.67, -0.09]$

gained 1.21% over B ($[0.70, 1.74]$, 240 episodes) and 4.79% over the alternative configuration A1 ($[2.01, 7.80]$), which itself gained 3.44% over B on the 160 non-cold episodes ($[1.78, 5.26]$); its longer-horizon check on 220 episodes found no persistent 30-day gain over B (-0.07%, $[-0.58, +0.44]$; 3.80% over A1, $[2.30, 5.22]$). The four-regime and five-day studies used weather-window bootstraps within class (percentile intervals). Offline four-regime selection gained 5.46% ($[4.75, 6.25]$). Over five days ($n = 80$) select-once and daily re-selection gained 4.34% and 6.30% over B ($[3.10, 5.57]$ and $[4.65, 8.35]$), daily over once 2.05% ($[-0.07, +5.09]$); over the nested 30 days ($n = 36$) they gave -5.45% and $+5.64\%$ ($[-14.24, +3.98]$ and $[3.50, 7.36]$), daily over once 10.52% ($[0.45, 18.66]$). The nested 30-day select-once result is sensitive to a single episode: removing one Fairbanks episode changes select-once versus B from -5.45% to $+3.01\%$ and the daily-versus-once ratio from $+10.52\%$ to $+1.17\%$.

Extreme-cold episodes. Thirteen extreme-cold 30-day episodes from six stations, all with windows used in earlier diagnostics, were evaluated separately: select-once gained 2.53% over B ($[1.91, 3.30]$, none of the 13 worse) and daily re-selection 1.93% ($[0.54, 3.59]$, one worse), while the alternative fixed profile A1 lost 5.73% ($[-7.43, -4.40]$, all 13 worse); no extreme-cold start occurs in the 2021 main sample. In a 67-episode first-workday comparison, the guarded extreme-cold profile gained 4.38% versus B, compared with -11.85% for the unguarded cold profile, but these episodes reused only 17 windows.

F Profiles and reproducibility

Algorithm 1: Supervisory profile reassignment (one episode)

Input: controller C, baseline B, library Φ, regime map r, schedule π

$\theta \leftarrow B$; *gated* $\leftarrow \varnothing$; *pending* $\leftarrow (\varnothing, 0)$;
for $t = 0, \ldots, 8639$ **do**
  if $t = t_d$ *for some day* d ***and*** *the window of 228 observations is complete* **then**
    $k \leftarrow r(\bar{T}_d)$;
    if $d = 1$ **then**
      $\theta \leftarrow \theta_k$;
      if $\pi = \mathrm{MG}$ **then** *gated* $\leftarrow$ (daily if $k = M$ else once);
    else if $\pi = \mathrm{PD5}$ ***or*** *gated* $=$ *daily* ***or*** *($\pi \in \{\mathrm{LIST}, \mathrm{WEEK}\}$ and d is an allowed day)* **then**
      if $\theta_k \neq \theta$ **then** $\theta \leftarrow \theta_k$;
    else if $\pi = \mathrm{TRIG2}$ **then**
      if $\theta_k = \theta$ **then** *pending* $\leftarrow (\varnothing, 0)$;
      else if *pending* $= (k, n)$ **then** *pending* $\leftarrow (k, n + 1)$;
      else *pending* $\leftarrow (k, 1)$;
      if *pending* $= (k, 2)$ **then** $\theta \leftarrow \theta_k$; *pending* $\leftarrow (\varnothing, 0)$;
  $u_t \leftarrow C(x_t, \theta)$; advance simulator;

The controller is the public rule-based K07 controller [3]. The baseline B overrides six of its settings (Table 16); all other settings are controller defaults. The mild profile is the one-hour pre-start alternative A1; the cold and hot profiles were chosen by separate searches (145 and 103 completed candidates); the mild search (89 candidates) retained the simpler A1 after a validation-tail criterion. The extreme-cold profile copies the cold profile with a modified water-scale parameter, introduced after poor severe-weather outcomes. Search allocations and acceptance criteria changed during de-

velopment, so the library is not the output of a single prospective optimisation protocol. Table 16 lists all overrides.

Table 16: Profile overrides of the baseline, the library profiles and the fixed comparators GL and G30. A dash denotes the controller default (not zero); † the default two-hour pre-start.

		Library profiles				Fixed comparators	
Parameter	B	E (guard)	C	M (=A1)	H	GL	G30
RAMP	0.25	0.3096	0.3096	0.25	0.1711	0.2866	0.3548
COLD_SAT	-0.8	-0.2481	-0.2481	-0.8	-0.8432	-0.9668	-1.0
KI	–	0.0058	0.0058	–	0.0051	0.006	0.0068
KP	0.075	0.0714	0.0714	0.075	0.0671	0.0663	0.0699
HEAT_U_SCALE	0.875	0.8628	0.8628	0.875	0.8640	0.9288	0.8747
PRESTART_H	2†	2	2	1	2	2	2
EVENING_STOP_H	–	0	0	–	0	0	0
NIGHT_HEAT_DROP_K	3	2.9775	2.9775	3	2.9772	3.2697	3.5886
MILD_SHIFT_K	0.75	0.8475	0.8475	0.75	0.5459	0.7442	0.7701
WATER_SCALE	–	1.0000	0.8842	–	–	1.0	0.9907
WARM_SAT	–	-0.7764	-0.7764	–	–	–	–
MIN_DAMPER	–	0.0130	0.0130	–	0.0728	0.0248	0.0
FORCED_SAT	–	–	–	–	-0.5923	–	–
FORCED_PRESSURE	–	–	–	–	-0.9813	–	–

Schedules. All schedules adopt the profile of the day-1 class; if the day-1 window is incomplete, B stays until a schedule's next permitted change, which never comes for PS5 and MG. LIST may reassign only on days 1, 2, 3, 5, 7, 14, 21 and 28, and WEEK only on days 1, 8, 15, 22 and 29. TRIG2 adopts a new class only after it appears in two successive complete windows; a window of the active class clears the pending evidence, while incomplete windows neither trigger reassignment nor erase it (no window was incomplete in the 2021 runs). A logged profile change is an actual change of the active profile after the initial assignment, not a decision check or an actuator event.

Fixed comparators. GL (best of the originally tested fixed profiles, identified retrospectively) and G30, selected after a budget-limited search of 64 candidates on 12 development episodes for the 30-day horizon, are fixed profiles; Table 16 lists their overrides. Both run from step zero.

Reproducing the primary estimate. From the per-episode results, keep only complete six-arm bundles and remove duplicate replay records (twelve rows duplicate two bundles; after removal there are 118 unique bundles: 98 main and 20 optional). Average each arm's loss over the 8,640 scored steps, compute the ratio of pooled losses to the pooled B loss with equal cell weights over the main episodes, and take the difference of baseline-referenced gains between MG and TRIG2. Resampling the 42 main-sample stations as described in Appendix E reproduces $+0.504$ pp $[+0.337, +0.704]$. The 20 optional episodes must not enter the primary estimate.

G Derivation of the switching threshold

An explanatory model for the regime dependence of Section 5.1 assumes that daily regimes follow a Markov chain in which regime k persists from one day to the next with probability P_{kk}, that running a profile matched to another regime i costs Δ_{ik} per day, and that each profile change costs c (transients and the risk of misclassification). Switching on entering regime k is then worthwhile only if

$$\Delta_{ik}\,\frac{1-P_{kk}^{H_d}}{1-P_{kk}} > c, \tag{6}$$

where the fraction is the expected remaining length of the spell within the H_d days left. With $c = 0$ this recovers daily reassignment; if all reachable spells are short it recovers select-once; and uncertainty in the observed regime favours waiting for confirmation, as persistence does. Spell length alone would favour frequent reassignment only after mild starts if spells that follow a mild start were long and departures from cold or hot starts brief; on the 2021 weather they are not (below).

Within the model, the gate's pattern then requires the daily cost Δ_{ik} of a mismatched profile to differ by regime, larger for a mild profile in cold or hot weather than for a cold or hot profile in mild weather; the post hoc split by day class below shows this asymmetry (Table 17), which was not measured in advance. Whether a rule that applies (6) directly can match the gate is an open question.

Proposition 1. *Let profile θ_i be active when a spell of regime $k \neq i$ begins with H_d days remaining, and let the spell length S be geometric with $\Pr(S \geq s) = P_{kk}^{s-1}$. Comparing a switch to θ_k at the start of the spell with keeping θ_i for the whole spell, switching has lower expected loss if and only if* (6) *holds. If switching is not worthwhile at the start of a spell, it is not worthwhile later in the same spell.*

Proof. The two policies differ only during the spell. Keeping θ_i incurs an extra loss Δ_{ik} on each of the $\min(S, H_d)$ days; switching incurs the one-off cost c. The expected difference is therefore

$$\Delta_{ik}\,\mathbb{E}[\min(S, H_d)] - c, \qquad \mathbb{E}[\min(S, H_d)] = \sum_{s=1}^{H_d} \Pr(S \geq s) = \sum_{s=1}^{H_d} P_{kk}^{s-1} = \frac{1 - P_{kk}^{H_d}}{1 - P_{kk}},$$

which is positive if and only if (6) holds. If the spell has already lasted m days, the remaining length is again geometric with the same parameter (memorylessness) and the remaining horizon is $H_d - m$; the same argument gives the condition with $H_d - m$ in place of H_d. Since the expected remaining length increases with the horizon, the condition is harder to satisfy later in the spell. □

The derivation ignores dependence between consecutive spells and the possible cost of switching back after a spell ends; including the latter would raise the effective switching cost.

Spells on the 2021 weather. From the class the controller observed each day in the 98 main episodes (the same in every arm), day-to-day persistence was alike for cold, mild and hot ($P_{kk} = 0.85$, 0.83 and 0.84), and spells in another class lasted a median of 2 days after every start (means 2.6, 3.4 and 4.2 days after cold, mild and hot starts; 29, 43 and 26 episodes), while the first spell lasted a median of 12, 5 and 4.5 days. The 20 extension episodes show the same. This check is post hoc.

Direction of the mismatch. Select-once and daily re-selection log the profile active after each daily decision, so each window between decisions runs either the same profile in both schedules or two different ones in a known direction; the loss differences of the windows add up exactly to that of the episode, and the two schedules were bit-identical before their first difference in all 98 episodes. By the class observed at the first decision (Appendix E uses the weather-file label; three episodes differ), daily re-selection beat select-once by 2.88 pp after mild starts and lost 0.45 and 0.48 pp after cold and hot starts. Table 17 splits these differences by the class of the day. Re-selection gained where it left the mild profile, on cold and hot days after mild starts, and lost where it entered it, on mild days after cold and hot starts. Windows with the same profile after a first difference, which carry the effect of earlier days through the building state, contributed at most 0.01 pp, and the four main directions had the same signs in the 20 extension episodes. Mild days were cooler within cold-start episodes (mean observed 15.1 °C) and warmer within hot-start ones (21.4 °C) than after mild starts (17.9 °C), near the edges of the class, where the profile of the neighbouring regime still fits; cold or hot days after a mild start are departures that the mild profile handles poorly. Re-selecting daily only after mild starts, as the gate does, keeps the gaining direction and avoids the losing one. The analysis is post hoc on the episodes that tested the gate and does not confirm it independently.

H Learning source: rules and parameters

Tables 18 and 19 describe four rules tested in different configurations: the learned optimal start and the pressure reset form the layer of the blind test; learned night precooling is the rule retargeted to Penguin; the unoccupied setup was tested on its own and as a third rule.

The values of each rule (Table 19) were fixed before its own held-out or blind test and were never tuned on test data. They are the defaults of the cited practice or engineering choices made once on development architectures. A mechanism passed its development gate if its mean gain over the composite development architectures was positive and no architecture was more than 1% worse.

Table 17: Daily re-selection against select-once on 2021 weather, by the class of each day (post hoc; classes as observed by the controller). Rate: loss of select-once minus that of daily re-selection on those days, in % of B's loss on the same days (positive: re-selection better); contribution: in pp of the baseline-referenced gain of the start class; 95% station-cluster intervals (3,000 replicates); same profile: days after a first difference with the same profile in both schedules; last column: rate in the 20 extension episodes.

Start	Day	Days	Episodes	Rate (%)	Contribution (pp)	Extension (%)
Mild	cold	158	21	$+11.14$ $[+9.12, +13.80]$	$+2.10$ $[+1.17, +3.14]$	$+7.84$
	hot	234	23	$+3.48$ $[+2.43, +4.41]$	$+0.77$ $[+0.38, +1.21]$	$+3.02$
	same profile	448	34	$+0.02$ $[-0.12, +0.15]$	$+0.01$ $[-0.04, +0.04]$	-0.08
Cold	mild	164	20	-6.47 $[-8.80, -4.31]$	-0.45 $[-0.76, -0.25]$	-7.80
	extreme cold	13	4	-0.08 $[-0.48, +1.72]$	-0.01 $[-0.05, +0.04]$	$+4.71$ (one episode)
	hot	1	1	-63.79 —	0.00 —	—
	same profile	279	24	0.00 $[-0.05, +0.07]$	0.00 $[-0.02, +0.02]$	-0.35
Hot	mild	283	23	-1.98 $[-3.16, -0.42]$	-0.47 $[-0.93, -0.10]$	-1.90
	same profile	257	21	-0.03 $[-0.07, +0.04]$	-0.01 $[-0.03, +0.01]$	-0.07

Table 18: The rules of the layer in one template (Section 3.3); bounds and parameters in Table 19.

Rule	Targets	Signal and trigger (learns)	Changes, within bounds	Returns control when	Needs from the controller
Learned start	comfort	morning zone deficit (time constant τ)	start of conditioning, lead in $[L^{\min}, L^{\max}]$	two consecutive mornings with a deficit above its prediction by a margin (for the episode)	a settable pre-start lead; zone temperatures; the occupancy schedule
Pressure reset	comfort	cooling requests of warm zones	duct static-pressure floor, never below the controller's command	no requests: trims back to its minimum, where it no longer overrides the controller; a per-unit guard (Table 19)	the duct static-pressure command; zone temperatures and cooling-loop saturation
Learned precooling	energy	warm nights with cool outdoor air (whether to precool, from the zone excess before opening)	unoccupied cooling limit	the night window ends; for the episode after two over-cooled mornings	a settable unoccupied cooling limit; outdoor temperature
Unoccupied setup	energy	outside conditioning hours	unoccupied cooling setpoint (32.2 °C)	conditioning starts	a settable unoccupied cooling setpoint; the occupancy schedule

The learning source runs on top of G30; the constants of B and G30 are in Table 16, and Algorithm 2 gives one step of the learning source.

I Out-of-family sets

All architectures were produced with the official scenario generator of the Smart Buildings Challenge [5]. The development set has six composite architectures and three of other shapes (trapezoid, T-shape, oval), each with two weather windows of the official training stations (Atlanta, Denver, Fairbanks, Phoenix, Singapore). The held-out set has four composite architectures and two of other shapes (H-shape, semicircle), each with two windows of a station not used in development (Minneapolis, starting on 8 January and on 25 June). Table 20 lists the blind set; OOD there means floor plans of other shapes from the same generator, not buildings from another distribution. The same 36 blind scenes were reused for every controller tested on them, and architecture-cluster resampling keeps the episodes of an architecture together but does not account for weather stations shared by several architectures. Seeds and configurations are in `experiments/03_out_of_family_and_held_out/` (development and held-out) and `scenarios/blind_set/` (blind).

Table 19: Parameters of the learning source.

Mechanism	Parameter	Value
Learned optimal start	target residual deficit $D^\star$; threshold for an informative morning	0.3 K; 0.3 K
	lead window $[L^{\min}, L^{\max}]$	$[2, 4]$ h ($L^{\max} = 3$ h until two informative mornings)
	initial time constant τ_0; smoothing of $\log \tau$; range of τ	1 h; 0.5; $[0.3, 3]$ h
	model check margin; freeze	0.5 K; after two consecutive misses
	controller's own lead (when the layer does not act)	2 h
Pressure reset (trim-and-respond)	setpoint range	$[200, 400]$ Pa
	request: own loop output on / off	≥ 0.95 / < 0.85; two requests if warmer than $T^{\text{hi}} + 1$ K
	trim per step; respond per request; maximum per step; ignored requests	12 Pa; 15 Pa; 32 Pa; 0
	rogue zone (its requests ignored for the episode)	requests for 24 consecutive steps with the floor at 400 Pa, on 2 days
	per-unit guard (overrides stopped for the episode)	overrides in $> 90\%$ of the conditioning steps on each of its last 3 weekdays, with no fewer requests on the last than on the first
Learned night precooling (second controller)	weather gate: trailing 24-h mean outdoor temperature	$\geq 11\,^\circ$C
	outdoor air below the zone mean by	2 K
	unoccupied upper limit on weekday nights 00–05 h	$\max(T^{\text{hi}} - 0.5,\ \max_z T_z - 0.25,\ T^{\text{lo}} + 0.5)$ K
	learned need: zone excess over T^{hi} when the controller's own pre-conditioning begins, smoothed over warm weekday mornings without precooling; threshold	smoothing 0.5; > 0 K
	safety: precooling off for the episode	after two over-cooled mornings

J Additional results for the learning source

This appendix follows the questions of Sections 5.2 and 5.3: the blind test in detail (Appendix J.1), how and when the gain arises (Appendix J.2), the headroom account (Appendix J.3), the other controllers and the checks of the mechanism (Appendix J.4), and where each rule paid (Appendix J.5). Tests registered before their outcomes say so in their headings and in the Status column of Table 4; everything else is post hoc and descriptive: it uses the frozen blind runs after the pre-specified analyses had been read and changes no pre-registered verdict.

J.1 The blind test in detail

Per architecture. Table 21 lists the gain of each contrast for each blind architecture and the recovery constant learned there (mean over its three episodes). Across the 12 architectures the gain followed the recovery constant estimated in the layer's own runs ($\rho = 0.81$, post hoc) rather than the static properties of the building (envelope heat capacity $\rho = -0.48$, facade U-value $\rho = -0.16$).

Tests. Table 22 reports, for every contrast on the 10 composite architectures, the two-sided bootstrap p-value, the Wilcoxon signed-rank and sign tests on the per-architecture gains, and the Holm

Algorithm 2: One step of the learning source (learned optimal start and pressure reset); values in Table 19, pressures in Pa.

Input: observation x_t; state $(\tau, n, \hat{D}, \textit{misses}, \textit{frozen}, f_a)$ and the rogue-zone and guard counters; opening time t^{o}

if *weekday* ***and not*** *started today* ***and not*** *frozen* ***and*** $t^{\mathrm{o}} - L^{\max} \le t < t^{\mathrm{o}} - L^{\min}$ **then**
  compute D_t (Equation (3)) and λ_t (Equation (4)), with $L^{\max} = 3\,\mathrm{h}$ before two informative mornings ($n < 2$);
  if $t^{\mathrm{o}} - t \le \lambda_t$ **then**
    set the controller's lead so that it starts conditioning now;
    if *the bound* $L^{\max}$ *did not cap the lead* **then** predict $\hat{D} \leftarrow D_t\, e^{-(t^{\mathrm{o}}-t)/\tau}$;
$u_t \leftarrow C(x_t, \theta_t)$; `// the unmodified controller acts`
for *each air-handling unit* a *not stopped by its guard, during conditioning hours* **do**
  if *first conditioning step of the day* **then** $f_a \leftarrow 200$;
  $R_a \leftarrow \sum_z m_z r_z$ over its zones, with $r_z \in \{0, 1, 2\}$ requests and $m_z = 0$ for a rogue zone;
  if $R_a > 0$ **then** $f_a \leftarrow \min(f_a + \min(15R_a, 32),\ 400)$ **else** $f_a \leftarrow \max(f_a - 12,\ 200)$;
  if *unit* a *runs* ***and*** $f_a > \max(u_{t,a}, 200) + 0.5$ **then** $u_{t,a} \leftarrow f_a$;
if *one hour after the controller started conditioning before opening, with* $D_s \ge 0.3\,K$ *at that start and* $0 < D_t < D_s$ **then**
  $\hat{\tau} \leftarrow 1\,\mathrm{h}/\ln(D_s/D_t)$; $\ln\tau \leftarrow \frac{1}{2}\ln\tau + \frac{1}{2}\ln\hat{\tau}$, clipped to $[\ln\tau^{\min}, \ln\tau^{\max}]$; $n \leftarrow n + 1$;
if *at opening, with a prediction* $\hat{D}$ **then**
  misses $\leftarrow$ *misses* $+ 1$ if $D_t > \hat{D} + 0.5\,\mathrm{K}$, else 0; **if** *misses* $= 2$ **then** *frozen* $\leftarrow$ true;
at noon: the controller's own lead $L^{\min}$ returns until the next morning;

Table 20: Blind set: 12 architectures, each with three 2022 weather windows whose first day is cold, mild or hot (station, start date, mean outdoor temperature of the first day in °C).

Seed	Group	Shape	Size (ft)	Floors	Zones	AHUs	Cold window	Mild window	Hot window
52200	composite	composite	618×318	2	29	7	Buffalo 13 Nov (1.7)	Tucson 13 Mar (18.2)	New Orleans 11 Sep (28.5)
52201	composite	composite	571×348	2	36	6	Spokane 6 Nov (3.7)	Salt Lake City 19 Jun (21.6)	Miami 4 Sep (31.3)
52203	composite	composite	406×348	3	32	6	Albuquerque 20 Nov (3.4)	Honolulu 27 Mar (22.7)	Salt Lake City 31 Jul (32.7)
52205	composite	composite	503×453	2	23	7	Kansas City 6 Feb (3.5)	Birmingham 20 Mar (15.0)	Orlando 9 Oct (26.1)
52206	composite	composite	608×365	2	70	7	Omaha 16 Oct (8.2)	San Diego 8 May (16.1)	Honolulu 20 Nov (27.1)
52207	composite	composite	634×499	4	76	8	Seattle 2 Jan (3.5)	Boston 25 Sep (18.0)	Oklahoma City 15 May (26.8)
52208	composite	composite	617×437	2	40	5	Pittsburgh 3 Apr (3.5)	Oklahoma City 11 Sep (20.2)	Raleigh 12 Jun (27.7)
52209	composite	composite	538×472	2	99	10	Chicago 17 Apr (3.8)	Charlotte 23 Oct (15.2)	Albuquerque 10 Jul (28.6)
52211	composite	composite	413×333	4	105	11	Billings 6 Feb (4.2)	Portland 22 May (18.5)	New York 21 Aug (25.6)
52212	composite	composite	449×451	3	38	4	Detroit 2 Jan (−6.0)	Indianapolis 15 May (22.4)	Tampa 26 Jun (30.2)
52300	OOD	triangle	519×477	3	28	3	Memphis 2 Jan (−0.6)	Pittsburgh 18 Sep (22.3)	Houston 31 Jul (31.6)
52301	OOD	parallelogram	674×311	2	35	5	Indianapolis 13 Nov (0.7)	Billings 28 Aug (20.5)	Boston 21 Aug (25.6)

Table 21: Gain (%) per blind architecture. τ: learned recovery constant (h). LNP: learned night precooling.

Arch.	Group	τ	G30 vs. B	layer vs. G30	learned vs. fixed	pressure reset	Penguin+LNP
52200	composite	1.45	+15.40	+2.25	+0.57	+1.34	+5.91
52201	composite	1.01	+6.18	+1.57	+0.77	+0.00	+1.27
52203	composite	1.14	+14.18	+2.16	+1.06	+0.66	+1.73
52205	composite	1.71	+13.70	+6.69	+4.52	+2.14	+2.08
52206	composite	0.82	+0.82	+0.02	+0.01	+0.00	+1.84
52207	composite	1.11	+10.33	+2.83	−0.00	+2.87	+3.50
52208	composite	1.40	+12.57	+4.09	+0.00	+4.07	+4.25
52209	composite	0.69	+2.27	−0.00	+0.00	+0.00	+1.34
52211	composite	0.60	+3.01	+0.55	−0.49	+0.00	+0.13
52212	composite	1.47	+6.46	+4.37	+1.69	+2.66	+0.85
52300	ood	1.39	+8.79	+2.73	−0.17	+2.88	+0.68
52301	ood	1.28	+5.71	+0.54	−0.03	+0.12	+3.79

adjustment of the bootstrap p-values over the six secondary contrasts (the primary contrast is the layer against G30). With 10 clusters the bootstrap p-values are approximate. The exact Wilcoxon test confirms the primary contrast ($p = 0.004$) and, after the same Holm adjustment, the pressure reset and Penguin's precooling (0.020 and 0.012), but not the three contrasts of the start (0.078–0.098), each of which rests on 8 of 10 architectures. The bootstrap resamples architectures and keeps all matched arms of each. In points of baseline-referenced gain the chain is additive: 8.62 from tuning, 0.28 from the fixed start rule, 0.86 from learning the recovery constant and 1.43 from the pressure reset.

Table 22: Tests on the blind set (10 composite architectures). Bootstrap: 3,000 resamples of the architectures with replacement, each drawn architecture entering with all its episodes as often as drawn. Holm (W): the same adjustment applied to the exact Wilcoxon p-values. LNP: learned night precooling. †: post hoc, made after a first version failed on this set.

Contrast	Gain (%)	p boot	p Wilcoxon	p sign	Holm (boot)	Holm (W)	better/worse
layer vs. G30	+2.82	< 0.001	0.004	0.021	—	—	9/1
G30 vs. B	+8.62	< 0.001	0.002	0.002	—	—	10/0
layer vs. B	+11.19	< 0.001	0.002	0.002	—	—	10/0
learned start vs. G30	+1.25	< 0.001	0.020	0.109	< 0.001	0.078	8/2
fixed start vs. G30	+0.31	< 0.001	0.027	0.109	< 0.001	0.082	8/2
learned start vs. fixed start	+0.95	0.023	0.049	0.109	0.023	0.098	8/2
layer vs. learned start	+1.58	< 0.001	0.004	0.004	< 0.001	0.020	9/0
weather-clustered bounds vs. layer†	+0.21	0.008	0.125	0.125	0.016	0.125	4/0
Penguin+LNP vs. Penguin	+1.88	< 0.001	0.002	0.002	< 0.001	0.012	10/0

Robustness of the primary contrast. Leaving out one architecture at a time kept the pooled gain of the layer between 2.39% and 3.15%; the mean per-architecture gain is 2.45%. Of the 36 episodes 29 improved, 3 lost and 4 were unchanged (the layer never acted); only one lost more than 0.1%: an OOD architecture in a cold January window, where 11 earlier starts improved comfort but cost heating energy and carbon before opening.

What to expect on a new architecture (post hoc). The intervals above are for the mean over these architectures; a new architecture from the same generator varies more. Counting, as in Table 3, every gain above zero, over the 10 composite architectures the layer gained on 9 with K07-G30 (the tenth changed by -0.001%), on all 10 with ZT Best, on 8 with Residual PPO (two lost more than 1%) and with Penguin's precooling on all 10. If the gain on a new architecture is exchangeable with these 10 (drawing it from the same generator and weather design makes this plausible but does not guarantee it), it falls between the smallest and the largest of them with probability at least $9/11$ over repeated samples, whatever the distribution (exactly $9/11$ when ties have probability zero): $[0.0, 6.7]\%$ for K07-G30, $[0.0, 11.1]\%$ for ZT Best, $[-1.4, 7.8]\%$ for Residual PPO and $[0.1, 5.9]\%$ for Penguin. The 30 pairs of a comfort controller and a composite architecture share 10 architectures and are not independent: 27 gained, as a description only. Counted by architecture, the independent unit, the layer gained under all three comfort controllers on 8 of the 10 (exact 95% interval 44–97%); on the other two, 52209 and 52211, Residual PPO lost 1.35 and 1.15%, and K07-G30 changed 52209 by -0.001%. This holds for this generator; other building distributions are untested.

Held-out episodes. Table 23 shows every held-out episode. The two composite architectures that lost did so only in the January window, where the learned start acted; in the June window the layer was idle and the change was zero. With two of the four composite architectures worse, the pre-specified rule failed on this set although the pooled gain was positive.

J.2 How and when the gain arises (post hoc)

Further post hoc diagnostics (the time constant learned in each episode, the ten worst days, the decisions by weather change) are in the repository's `analyses/` folder.

Time to value. Each blind episode starts with no knowledge of the building; Figure 6 shows the gain over G30 up to each day and per week over the 30 composite episodes (post hoc). The fixed start gained 0.22% by day 7 and 0.31% by day 30, the learned start 0.41, 1.09 and 1.25% by days

Table 23: Held-out episodes (layer vs. G30; Minneapolis windows starting on 8 January and 25 June): gain (%), learned-start extensions, share of occupied steps with a pressure reset (%), and whether the safety freeze triggered.

Architecture, window	Group	Gain	Extensions	Reset (%)	Freeze
52001, January	composite	+1.23	11	0.07	yes
52001, June	composite	+0.00	0	0.18	no
52003, January	composite	+6.46	22	5.17	no
52003, June	composite	+11.10	0	34.95	no
52006, January	composite	−0.14	4	0.00	no
52006, June	composite	+0.00	0	0.00	no
52009, January	composite	−0.34	7	0.00	no
52009, June	composite	+0.00	0	0.00	no
52102, January	OOD	+0.71	6	0.00	yes
52102, June	OOD	+0.08	0	0.45	no
52108, January	OOD	+0.48	15	0.00	no
52108, June	OOD	+8.65	0	15.86	no

7, 14 and 30; its lead over the fixed start, in points of G30's loss, was +0.19 [+0.02, +0.43], +0.80 [+0.18, +1.39] and +0.94 [+0.10, +1.82], above zero from day 6, as the time constant came within 10% of its end-of-episode estimate only after a median of 11 informative mornings (75th percentile 15; 28% of episodes by the fifth; learning engaged in all 36 episodes, with a median of 13 informative mornings). The pressure reset, a feedback rule with no learned constant, paid from the first workday: with it the layer was 0.67 points above the learned start alone on day 2 and 2.86 by day 7, and gained 3.28% by day 7 and 2.82% by day 30.

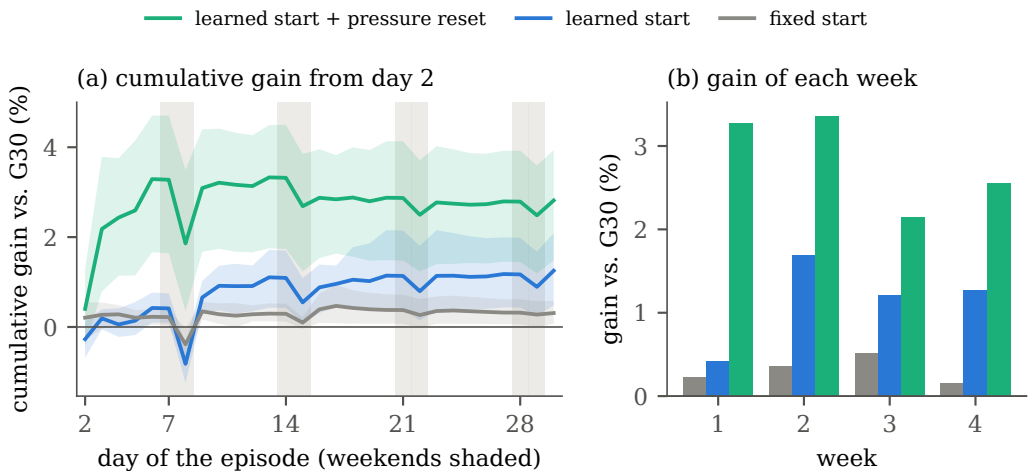


Figure 6: Time to value on the blind set (post hoc; 30 composite episodes, each starting with no knowledge of the building). (a) Gain over G30 accumulated up to each day, with 95% architecture-cluster intervals; weekends shaded. Day 1, a Sunday that ends with the first pre-start, which every start variant begins earlier than G30, is −13.8% for all three and is omitted; the dips on later Sundays have the same cause. (b) Gain of each week.

Days and hours. Figure 7 shows the gain per calendar day by outdoor temperature and by its change from the previous day, and the change of each loss component by hour. The gain was 4.1% and 4.2% on the coldest and hottest 10% of days, against 2.1% on the others; the layer pays a little energy at 02:00–03:00 to start earlier. Of the 900 composite episode-days, 52% improved and 2.6% lost more than 1% (5th and 95th percentiles −0.23% and +8.9%); most are unchanged because the layer acts only when there is something to correct. Of the ten worst days, five are Mondays and six are US public holidays (2022-01-17, 02-21, 07-04 twice, 10-10 and 11-24), without occupants in the simulator (summed occupancy zero, against 1,840–11,388 on the following day): neither the controller nor the layer can know a holiday, and the earlier start adds energy that buys no comfort.

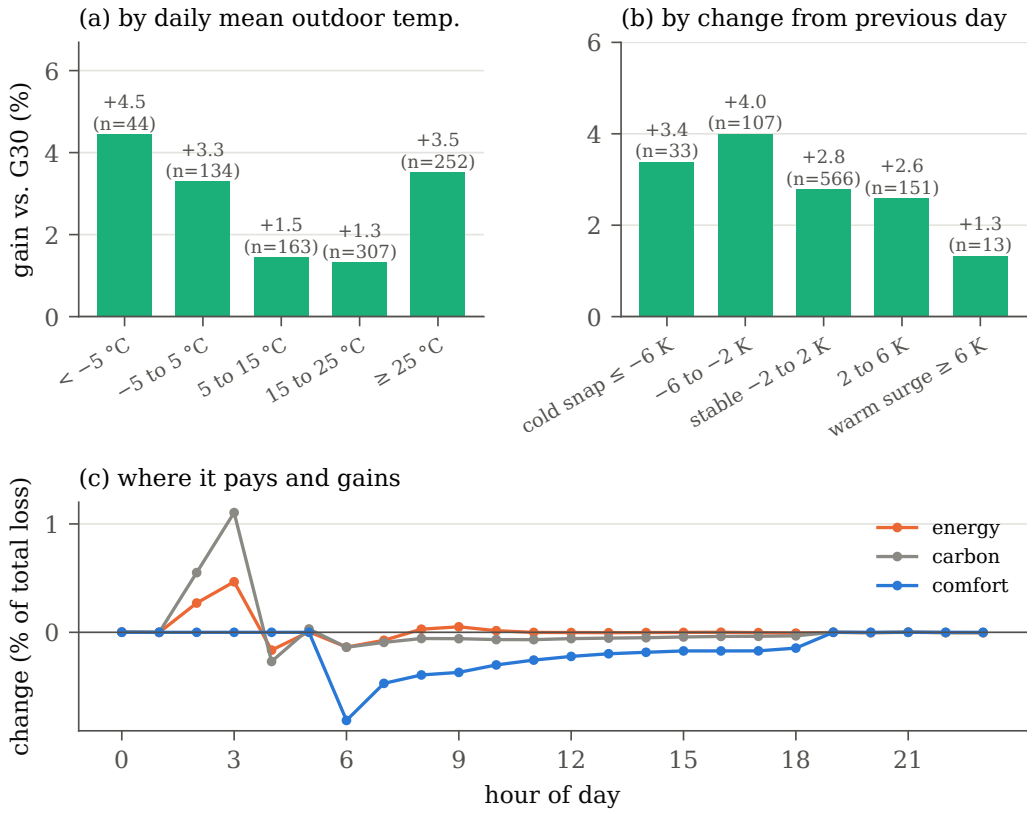


Figure 7: Gain of the learning source over G30 on the blind set, per calendar day: (a) by daily mean outdoor temperature, (b) by change of the daily mean from the previous day; (c) change of each weighted loss component by hour of day, in % of the total loss of G30. Panel (b) has no value for the first day of an episode; post hoc and descriptive, and days are not independent units.

The worst, Presidents' Day 2022 in Kansas City (-16.14%), followed a weekend whose daily mean rose from -2.8 to $+7.9\,^{\circ}\mathrm{C}$: at 02:00 the slowest blind building had a deficit of 4.4 K and a learned time constant of about 2 h, so the rule asked for a 5.3-h lead and started at the 4-h bound; nobody came, and the small loss of an unoccupied day makes the relative loss large.

Decisions and weather surprise. The start is decided at night without a forecast: in 87% of 122 decisions the predicted deficit at opening was within 0.5 K of the actual one, and even on the mornings it missed, the day was better than with G30. 17 further starts reached the lead bound and are not judged.

Where the gain comes from. Table 24 splits the change of the weighted loss components into occupied and unoccupied steps. Learned precooling on Penguin shifts some cooling to the night and saves more during the day, whereas on K07 the layer buys comfort with a small energy cost before opening.

Table 24: Change of the weighted loss components, in % of the comparator's total loss (blind set, composite).

	Energy		Carbon		Comfort
Contrast	occupied	unoccupied	occupied	unoccupied	occupied
K07: G30 + layer vs. G30	-0.07	$+0.50$	-0.70	$+1.32$	-3.87
Penguin + LNP vs. Penguin	-2.28	$+0.82$	-0.54	$+0.18$	-0.06
K07: G30 + setup vs. G30	$+1.11$	-2.29	$+0.25$	-0.54	$+2.09$
K07: setup + learned start vs. setup	-0.68	$+0.77$	-0.85	$+1.36$	-2.48

J.3 The headroom account

Headroom. Write the loss of an episode as $L = \sum_c w_c \ell_c$ over energy, carbon and comfort, let T and O be the weighted losses of the terms a mechanism targets and of the others ($L = T + O$), and pool over a set S of episodes. The gain of the layer l over its comparator b is then exactly

$$g_S = B_S - C_S, \qquad B_S = \kappa_S \, h_S \le h_S, \tag{7}$$

with the targeted share $h_S = 100 \sum_S T_b / \sum_S L_b$, the part of the targeted loss removed $\kappa_S = \sum_S (T_b - T_l) / \sum_S T_b \le 1$, and the cost in the other terms $C_S = 100 \sum_S (O_l - O_b) / \sum_S L_b$; hence $g_S \le h_S$ whenever $C_S \ge 0$, and g_S follows h_S when κ_S and C_S vary less than h_S. Table 25 gives each term (post hoc; the neural controller with the same checks as the others) and checks the relation of Section 5.3. Gain and share both divide by the comparator's loss, but a permutation that keeps only this common denominator gives correlations near zero, below the interval of each observed one, and for the comfort layer the relation also holds over architectures, without ratios and with one architecture left out (Table 25). The layer removed 15–24% of the comfort loss on all three comfort controllers; on K07-G30 and ZT Best it removed a larger part where the share was larger, while on Residual PPO the part removed did not rise with the share ($\rho = -0.30$), so there the association comes from the size of the headroom, and the cost in the other terms decided the two architectures that lost (Appendix J.4). The same architectures left comfort loss under all three ($\rho = 0.82$–0.94 between their per-architecture shares), and the layer gained on the same ones ($\rho = 0.65$–0.71 between their per-architecture gains; with Penguin's precooling 0.13–0.52). Penguin's precooling targets terms that make up almost all of its loss, so that share bounds little; it removed under 2% of it, and its relation with the share is weak over architectures and without ratios. All of this is post hoc on the frozen blind runs. The predictions registered for ZT Best and Residual PPO reuse the same 36 blind scenes: they test whether the account transfers across controllers, not whether it holds on new buildings.

Table 25: Decomposition of the blind-set gains and checks of their relation with the share of the targeted terms (post hoc). Top: in % of the comparator's loss, pooled over the 10 composite architectures (Equation (7)); a positive change of the other terms is a cost. Bottom: Spearman correlation of the episode gain with the share of the targeted terms; the null permutes the loss reductions across episodes (10,000 permutations), which keeps only the denominator that gain and share have in common; without ratios: absolute loss reduction against the absolute targeted loss; one architecture left out: the correlation over the episodes recomputed without each architecture in turn.

	K07-G30 + layer	ZT Best + layer	Residual PPO + layer	Penguin + precooling
Targeted terms	comfort	comfort	comfort	energy, carbon
Targeted share, h	15.9	44.2	22.4	97.2
Benefit, B	3.87	6.62	4.70	1.82
Cost in the other terms, C	$+1.05$	$+0.91$	$+1.34$	-0.06
Net gain, g	2.82	5.71	3.37	1.88
Part of it removed, κ	24.3	15.0	21.0	1.9
ρ over the 36 episodes	$+0.80$	$+0.93$	$+0.75$	$+0.68$
95% interval, by architecture	$[+0.57, +0.89]$	$[+0.78, +0.97]$	$[+0.51, +0.87]$	$[+0.53, +0.81]$
common denominator alone, mean	-0.09	-0.18	-0.12	-0.04
95% range	$[-0.42, +0.25]$	$[-0.49, +0.14]$	$[-0.44, +0.21]$	$[-0.33, +0.25]$
ρ over the 12 architectures	$+0.78$	$+0.92$	$+0.78$	$+0.30$
ρ without ratios	$+0.84$	$+0.95$	$+0.75$	$+0.28$
ρ, one architecture left out	$+0.77$ to $+0.83$	$+0.91$ to $+0.94$	$+0.72$ to $+0.78$	$+0.65$ to $+0.73$
ρ(part removed, share)	$+0.54$	$+0.60$	-0.30	$+0.68$

Early indication. The share in Table 25 is measured over the whole episode. To see whether a short window indicates it in advance, three indicators were fixed from the first week (days 1–7) of G30 before this analysis was first run (exploratory, post hoc on the blind runs): the comfort share of that week's loss, the mean distance of the zone temperatures at each opening from their median over the occupied hours that follow, and the time from the opening until the mean zone temperature comes within 0.5 K of that median; the last two use sensors only. Table 26 relates them to the gain over days 8–30, which does not overlap the window. The first-week comfort share ranked the layer's later gain almost as well as the share of the whole episode ($\rho = 0.72$ against 0.77; 3.95% against

0.68% above and below its median), whereas the two temperature indicators did not; for the learned start alone, the temperature distance at opening split the later gain at its median (2.48% against 0.05%), but its rank correlation was weak. With 10 architectures and the runs that measured the layer, this is an indication for further study, not a validated selector; using it before the layer acts would also need a first week without the layer to measure that share.

Table 26: Early indicators from the first week of G30 and the gain over days 8–30 (post hoc, exploratory): Spearman ρ over the 30 composite episodes with 95% architecture-cluster intervals, its range when one architecture is left out, and ρ over the 10 architectures. The last row uses the whole episode and is not an early indicator.

	Layer			Learned start
Indicator	ρ [95%]	one left out	architectures	ρ [95%]
Comfort share of the loss	$+0.72$ $[+0.24, +0.88]$	$+0.64$ to $+0.84$	$+0.82$	$+0.19$ $[-0.18, +0.46]$
Temperature distance at opening	$+0.23$ $[-0.09, +0.53]$	$+0.14$ to $+0.32$	$+0.55$	$+0.38$ $[-0.03, +0.71]$
Recovery time after opening	$+0.13$ $[-0.20, +0.43]$	$+0.06$ to $+0.26$	$+0.41$	$+0.08$ $[-0.17, +0.29]$
Comfort share, whole episode	$+0.77$ $[+0.38, +0.90]$	$+0.71$ to $+0.85$	$+0.82$	$+0.44$ $[+0.18, +0.63]$

The objective. This trade pays because comfort carries 60% of the weight: re-weighted with comfort weight w and $(1 - w)/2$ on energy and on carbon, the same runs give 0.26% with equal objective weights and a loss below a comfort weight of 0.3 (Figure 3b); learned precooling on Penguin keeps 1.9–2.3% across this sweep.

J.4 Other controllers and checks of the mechanism

Each test marked registered in this subsection was frozen in a protocol before its runs (Table 12).

A trained neural controller (registered). The public Residual PPO controller [26] runs a tuned per-zone PI rule of the ZT Best lineage with an off-hours rule, plus a residual policy trained with PPO in a chain of runs on official training buildings and, in the last run, on 2,700 generated buildings, so the blind architectures are new to the layer but not necessarily outside the network's training distribution. It has one shared network per entity type (zone, air handler, plant; two hidden layers of 128 units), copied to every entity, so that the same weights run any building; each copy reads its own entity's features and fixed summaries of the entities connected to it, without learned message passing. The layer was attached unchanged: it reads and sets the rule inside the controller, and its pressure floor applies to the final command, after the network's correction. Under a protocol frozen before any blind run (with the layer switched off the runs were bitwise identical to the controller's own; with it on, identical up to its first action), it gained 3.37% $[1.57, 5.22]$ over the 10 composite architectures, 8 of 10 better (sign test $p = 0.11$: evidence of an aggregate gain more than of a gain on every architecture) and +5.58% on the OOD pair; two architectures lost 1.15 and 1.35%, so the pre-specified rule (none more than 1% worse) failed. The gain was a comfort benefit of 4.70 points against energy and carbon costs of 0.61 and 0.73; the controller left 22% of its pooled blind-set loss in comfort. Of four predictions registered before the outcome, three held: the gain rose with the controller's comfort share ($\rho = 0.75$), the gain over all 36 episodes (3.78%) exceeded K07's 2.69% as its mean per-episode comfort share (21%) exceeded K07's 13%, and the first-week share ranked the gain over days 8–30 ($\rho = 0.51$); the lowest-share third gained 0.70%, outside $\pm 0.5\%$. On six official training scenes, from buildings and climates the network was trained on, where it leaves 2.5% of its loss in comfort, the layer gained 0.14%.

Where the neural controller gained and lost (post hoc). Across the 12 architectures, rather than the 36 episodes of the registered test, the gain also followed the controller's comfort share (Spearman $\rho = 0.78$). The two architectures that lost differed: on 52209 the controller left 3.0% of its loss in comfort, so the layer removed only 0.25 points while energy and carbon rose by 1.60; on 52211 it removed 1.87 points of comfort loss, but energy and carbon rose by 3.0, from 7.5 more at night (00–05 h) only partly recovered in the morning. On the six official training scenes the pattern was the same: night-time energy and carbon rose on all six, and comfort improved clearly only on the two scenes where the layer gained. On the leaderboard the layer worsened the controller's score

from 0.01053 to 0.01064 (-1.0%), against the registered prediction of no change, which rested on six official scenes, in sample for the network, on four of which the layer had already lost.

An early warning (exploratory). Both architectures on which the layer lost had no gain in the first week (-0.05 and -2.41%); so did one that later gained (52301, -0.10%, then $+3.40\%$ over days 8–30), whereas the first-week comfort share, which ranked the later gain on K07, flagged only one of the two. A layer that switches itself off after a first week without gain would thus have avoided both losses at the cost of that later gain. The analysis is retrospective: a deployable rule would need an observable proxy of the first-week gain and a test on new architectures.

A graph network imitating the neural controller (registered). A graph network over zones, air handlers and the plant (two rounds of message passing) was trained to imitate the residual policy of the neural controller on the held-out and official training buildings; on four validation episodes from other buildings it reproduced it closely ($R^2 \geq 0.97$ for every output; loss 0.9998 of the teacher's). It therefore tests the unchanged layer on a graph-structured implementation of the same policy, not on an independent controller. With all 72 blind runs carrying the registered weights, the unchanged layer gained 3.55% $[1.82, 5.40]$ over the 10 composite architectures, 8 of 10 better and +5.73% on the OOD pair; 52209 lost 1.17%, so the pre-specified rule failed, as on the neural controller there. The gain was a comfort benefit of 4.75 points against energy and carbon costs of 0.57 and 0.64. All four predictions registered before the outcome held: the gain rose with the controller's comfort share ($\rho = 0.74$), the lowest-share third gained 0.48%, within $\pm 0.5\%$, the gain over all 36 episodes (3.96%) exceeded K07's 2.69% as its mean per-episode comfort share (22%) exceeded K07's 13%, and the first-week share ranked the gain over days 8–30 ($\rho = 0.49$). Alone, the graph network was 0.80% worse than the controller it imitates ($[-1.94, +0.21]$).

Which rule brings the gain on the neural controller (registered). Each rule of the layer was run alone on the neural controller's 36 blind scenes, on a different machine from the controller alone and the full layer (such runs differ across machines by at most 10^{-6} per step). The pressure reset alone gained 2.99% $[1.60, 4.53]$ over the 10 composite architectures, all 10 better (+5.07% on the OOD pair), and the learned start alone 0.49% $[-0.10, 1.03]$, 7 of 10 better; together, 3.37%. All four registered predictions held: each rule alone gained, both together gained at least as much as the better one, and each lowered the comfort loss (by 1.46 and 3.37 points; together 4.70). Both losses of the layer came through the start: alone it lost 1.34% on 52209 and 1.15% on 52211, where the reset alone gained, and added to the reset it gained 0.39% $[-0.17, 0.92]$ on average but lost up to 1.41%. Its earlier starts, before opening, raised energy and carbon (0.33 and 0.64 points), which on these two architectures the comfort they bought did not repay; the reset alone would have passed the layer's pre-specified rule.

Learning inside the full layer (registered). The layer with the start's time constant fixed at 1 h was compared with the full layer, both with the pressure reset, on the 36 blind scenes. Learning added 0.94% $[0.10, 1.82]$ over the 10 composite architectures (7 of 10 better, exact Wilcoxon $p = 0.084$; -0.18% on the OOD pair), so by its registered pooled criterion learning adds value inside the layer; it removed 1.50 points of comfort loss at 0.56 more in energy and carbon. In points of baseline-referenced gain, learning added 0.84 with the reset and 0.86 without it, and the reset added 1.45 on the fixed start and 1.43 on the learned one: the two parts add up, and neither absorbs the other. As in the start contrasts above, the evidence is stronger for the pooled gain than for every architecture. Two episodes of the full layer re-run on the second machine matched their originals bit for bit.

The value of knowing the time constant early (registered). Given each episode's final time constant from the first day (an oracle, not deployable), the layer gained 0.02% more ($[-0.003, +0.057]$; 6 of 9 changed composite architectures better), 0.044% over days 1–14 and 0.010% over days 15–30; both registered predictions held. This is the value of early knowledge of τ for this start rule, not an upper bound on what a different rule could gain.

A third rule (registered). The Guideline 36 unoccupied setup, added to the learned start and the pressure reset with its parameter as in the unoccupied-setup test, gained 0.38% over the layer ($[-0.45, +1.49]$; 5 of 10 composite architectures better, the worst -2.46%): it removed 1.29 points of energy and 0.32 of carbon and added 1.23 of comfort loss. Both registered predictions held:

adding the setup had the same sign as adding it on top of the learned start alone, and it lowered energy and carbon.

J.5 Where each rule paid: official scenes, leaderboard and an unoccupied setup

Table 27: Kaggle public leaderboard (lower is better). Scores were never used to choose a configuration.

Submission	Score	Submission	Score
K07, baseline B	0.01517	G30 + learned start	0.01477
K07, one-time gate (MG)	0.01549	G30 + layer (start, pressure reset)	0.01477
K07, tuned fixed G30	0.01477	G30 + unoccupied setup	0.01414
Penguin (reproduced)	0.01004	G30 + unoccupied setup + learned start	0.01415
Penguin + learned precooling	0.01001	Residual PPO, alone / + layer	0.01053 / 0.01064

Table 28: Which source paid where (summary of Section 5). Gains pooled over the composite architectures; P: pre-specified, R: registered in advance, D: descriptive.

Situation	Source	Result	Status
Buildings of the profiles' own families	one-time profile gate	$+0.504$ pp over persistence; registered replication $+0.255$ pp, inconclusive	P, R
New buildings	frozen profiles	did not transfer (held-out -0.9%; leaderboard worse than B)	P, D
New buildings; controller leaves comfort loss (mean share 13–36%)	learned start + pressure reset	$+2.82\%$ (K07-G30), $+5.71\%$ (ZT Best), $+3.37\%$ (Residual PPO; per-architecture rule failed)	P, R
New buildings; controller loses mainly in energy and carbon (97%)	learned precooling	$+1.88\%$ (Penguin), all 12 architectures improved	P
Little loss left in comfort (official scenes, 0.2–4.6%)	learned start + pressure reset	leaderboard unchanged (K07), 1.0% worse (neural controller)	D, R

Where each rule paid. Table 29 collects the tests of the learning source, on unseen architectures, on the official scenes and on the leaderboard. The comfort rules gained where the controller left comfort loss and not where it left almost none; the energy rules gained where energy could be saved at little cost in comfort. Only the tests marked R were predicted in advance; the account of the others is post hoc.

Official training scenes. On the official training buildings the pressure reset of K07-G30 acted in 19 of 2,880 steps and the learned start only after cold nights, consistent with K07's unchanged five-decimal leaderboard score, though the activity on the hidden scenes is unknown. In the warm official scenes K07 spends 9–25% of its loss on energy during unoccupied hours, because it follows the official unoccupied band. The Guideline 36 unoccupied cooling setpoint (32.2 °C) passed its gate (development $+1.26\%$, official scenes $+2.25\%$), saving 11.4% on the hottest official scene and nothing where nights are cool (per scene, with and without the learned start, in Table 30). On that scene, in the first three occupied hours of weekdays, the zone furthest from the band was 0.57 K outside it on average (0.22 K without the setup) and the mean zone 0.04 K, and the learned start changed nothing (one scene). Learned precooling acted in only one of eight official training scenes, where it reduced Penguin's loss by 2.7% over ten days, so it was expected to leave Penguin's score nearly unchanged.

Unoccupied setup on the blind set. The setup, the setup with the learned start and the learned start alone were compared with G30 on the blind set under a protocol fixed before any setup run on this set was read (Table 31). Its rule for the combination against G30 (lower composite loss, most changed composite architectures better, none more than 1% worse) failed on the last condition and held against the setup. By term (Table 24), the setup saved unoccupied energy and carbon but cost more during occupancy, mostly comfort, which the learned start more than recovered; found by a diagnosis of the official scenes, its night saving did not cover its daytime cost on the blind set.

Table 29: Where each rule paid: the tests of the learning source by controller and setting. Gains are pooled loss reductions over the episodes of the set (leaderboard: change of the public score); P: pre-specified test; R: outcome registered in advance (as in Table 4).

Controller	Rule(s), target	Scenes	What the controller left	Gain (%)
K07-G30	start + reset, comfort	unseen (P)	comfort 13% of its loss	+2.82
ZT Best	start + reset, comfort	unseen (P, R)	comfort 36%	+5.71
Residual PPO	start + reset, comfort	unseen (R)	comfort 21%	+3.37
Graph network (imitating it)	start + reset, comfort	unseen (R)	comfort 22%	+3.55
Residual PPO	reset alone, comfort	unseen (R)	comfort 21%	+2.99
Residual PPO	start alone, comfort	unseen (R)	comfort 21%; night-time cost on two architectures	+0.49
Residual PPO	start + reset, comfort	official training scenes	comfort 2.5% (in sample for the network)	+0.14
K07-G30	start + reset, comfort	leaderboard	comfort 0.2–4.6% on the official scenes	0.00
Residual PPO	start + reset, comfort	leaderboard (R: no change)	comfort 2.5% on the official scenes	−1.04
Penguin	precooling, energy	unseen (P)	energy and carbon 97%	+1.88
Penguin	precooling, energy	leaderboard	acted on 1 of 8 official scenes	+0.30
K07-G30	unoccupied setup, unoccupied cooling	official scenes; leaderboard	9–25% unoccupied energy; acted on the two hot scenes	+2.25; +4.3
K07-G30	unoccupied setup, unoccupied cooling	unseen (P, R)	saved 2.8 points, cost 3.5 (mostly comfort)	−0.62
K07-G30 + layer	unoccupied setup added	unseen (R)	over the layer	+0.38

Table 30: Unoccupied setup on six official training scenes: gain (%) of the setup over G30 and of the learned start on top of it, mean night outdoor temperature (°C) and the highest mean zone temperature when unoccupied (G30 / setup, °C).

Scene	Setup vs. G30	Learned start added	Night T_{out}	Max zone T unoccupied
building_train_1 / denver_7	−0.00	−0.00	11.5	22.9 / 22.9
building_train_1 / fairbanks_1	−0.00	+0.08	−21.0	22.8 / 22.8
building_train_1 / phoenix_5	+11.44	−0.00	35.9	25.4 / 31.8
building_train_1 / singapore_8	+1.97	−0.00	26.3	25.0 / 27.1
building_train_2 / fairbanks_2	−0.00	−0.00	−9.6	22.7 / 22.7
building_train_4 / phoenix_8	−0.00	−0.00	13.4	23.1 / 23.1

Table 31: Unoccupied setup on the blind set, under a protocol fixed before any setup run on this set was read: gain (%) pooled over the 10 composite architectures, composite architectures better of those changed, the lowest architecture gain, and the pooled gain on the two OOD architectures. U: setup; R: learned start; UR: both.

Contrast	Gain	Better	Worst	OOD
U vs. G30	−0.62	3/10	−5.14	−2.95
R vs. G30	+1.25	8/10	−0.16	+0.00
UR vs. U	+1.87	10/10	+0.01	+0.77
UR vs. G30	+1.27	7/10	−3.31	−2.17

Variants that did not pass. The K07 start attached to Penguin's pre-conditioning time lost 0.67% in development; a first weather-clustered version of the start bounds (longer lead in deep cold) lost 0.36% on the blind set because the safety freeze triggered in two slow buildings, and the corrected version, made after this failure and run on the same blind set, gained 0.21% (post hoc; Table 22).

K Compute, software and licences

All runs used CPUs only: a laptop (AMD Ryzen 7 5800H), a shared workstation and Kaggle CPU notebooks; the same K07 episode of the learning source gave bitwise-identical results on different machines; the hosted runtime of the profile study passed a tolerance screen instead (Appendix D), and the neural controllers differ across machines by at most 10^{-6} per step in the reward. A 30-day episode took a median of 21–49 minutes on one core, depending on the machine, and about 1.2 GB of memory; the controllers needed 1–4 ms per step, so the simulator dominates. The simulator is the evaluation kit of the Smart Buildings Challenge [5], identified by the SHA-256 of its evaluation script (`947c968ce3a891da8334b7504ac0e7b45bd14fe13400f0136df49825d5247792`); every run records that hash and the hashes of its runner, controller code and scenario configuration, and most record the NumPy version (2.0 or 2.1). K07, ZT Best, Penguin (which builds on a public controller by Daniel Wu [31]) and the Residual PPO controller are public notebooks under the Apache License 2.0 [3, 24–26]; weather comes from NOAA's Integrated Surface Database [23]. The competition data are licensed CC BY-NC-SA 4.0. Bootstrap and permutation tests use fixed seeds.

Reproducing the learning-source result. Average the absolute reward of G30 and of the layer over the 8,640 steps of each blind episode, pool both over the 30 episodes of the 10 composite architectures and compute $100\,(1 - \sum L_{\mathrm{layer}} / \sum L_{\mathrm{G30}}) = 2.82\%$; resampling the composite architectures 3,000 times with a fixed seed gives $[1.48, 3.94]$.